A multivariate Bayesian framework for diachronic dating of ancient texts: Validation on Biblical Hebrew and Ancient Greek

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Abstract

The linguistic dating of ancient texts from internal morphosyntactic evidence is central to biblical studies and classical philology, but existing approaches assign ordinal labels rather than calibrated probability distributions, leaving claims about relative chronology without quantified uncertainty. The present paper introduces a Bayesian framework for diachronic dating and applies it to Pentateuchal source chronology and the authenticity of embedded archaic poetry. The core model (MLE-MVN) combines per-feature OLS regression with a Tikhonov-regularized multivariate normal likelihood over a fine date grid; a companion hierarchical model (HB-VI) adds group-level shrinkage and variational inference for better-calibrated credible intervals. A resistant model restricted to four clause-level syntactic features below conscious editorial control yields an archaism diagnostic from the divergence between full and resistant MAP dates, and a genre correction downweights features sensitive to legal versus narrative register. Applied to 22 texts from the BHSA database spanning 760–167 BCE, the models place the Priestly source at $\sim$402 BCE under agnostic priors, a data-driven result (prior sensitivity Class D) suggesting linguistic crystallization well after the traditionally assumed Exilic period, and the Deuteronomic Code at $\sim$698–737 BCE, consistent with an Iron II origin predating the Josianic reform. The archaism diagnostic identifies the Song of the Sea (Exod 15) as a late archaizing composition (resistant MAP $\approx$ 460 BCE, $\Delta_{\text{arch}} = +392$ yr) rather than an ancient text; the Song of Moses (Deut 32) converges to a nearly identical resistant date. Held-out oracle Jeremiah chapters return MAP = 562 BCE, consistent with seventh-century authorship, and cross-language validation on a 63-text Ancient Greek corpus recovers five held-out texts within $\sim$30 yr of established scholarly dates. Code and corpora are publicly available (see Data Availability).

Introduction

Scholars in biblical studies, classical philology, and Near Eastern linguistics routinely use internal linguistic evidence to constrain the dates of ancient texts. The transition from Classical Biblical Hebrew (CBH; also called Standard Biblical Hebrew, SBH) to Late Biblical Hebrew (LBH) is among the best-documented diachronic shifts in any ancient language [1–3], and comparable transitions are well attested in Ancient Greek [4] and Akkadian. Despite this rich empirical tradition, linguistic dating almost never produces calibrated probability estimates; scholars identify features as "archaic" or "late" and assign texts to qualitative categories, but the resulting claims carry no quantified uncertainty.

A notable recent contribution by Faigenbaum-Golovin et al. [5] demonstrated that statistical analysis of lemma n-gram frequencies can reliably distinguish among the three major scribal corpora of the Hebrew Bible — the old layers of Deuteronomy (D), the Deuteronomistic History (DtrH), and the Priestly source (P) — achieving approximately 84% chapter-level attribution accuracy on a 50-chapter benchmark, with interpretable discriminating-lemma lists linking computational results to traditional philological categories. That study establishes the viability of quantitative computational methods for biblical Hebrew scholarship and demonstrates that linguistically distinct scribal schools leave detectable statistical fingerprints even in relatively short texts. Its focus, however, is *authorship attribution*: given a passage of disputed origin, which known scribal milieu produced it? The present paper addresses the complementary but methodologically distinct question of *diachronic dating*: not "which school wrote it?" but "when was it written?" These are related but separable questions that demand different statistical frameworks, different feature sets, and different validation strategies, as detailed below.

This paper addresses two methodological gaps that authorship-attribution approaches leave open. First, no existing probabilistic framework provides date posteriors with credible intervals for Biblical Hebrew texts. Qualitative assessments of individual features are genuinely informative, but they cannot be combined into a single coherent posterior without a model that propagates feature-level evidence into a joint probability distribution over dates. Second, no principled framework separates the temporal signal from two well-known confounds: *archaism* (a late composer deliberately employing archaic vocabulary and morphology) and *genre* (legal or narrative register producing feature distributions that mimic either earlier or later stages of the language independently of composition date).

The framework presented here makes the following contributions:

1. A morphosyntactic feature extraction pipeline operating on the BHSA/ETCBC database [6,7], producing 35 features across three tiers (lexical, morphological, and clause-level syntactic), selected by Spearman correlation ($p < 0.30$) with 100% LOO sign-consistency.

2. The MLE-MVN Bayesian dating model: per-feature OLS regression, Tikhonov-regularized residual covariance, and posterior over a 500-point date grid with explicit uncertainty quantification.

3. The HB-VI model: hierarchical Bayesian regression with mean-field variational inference (MFVI) and ADAM optimization [8,9], providing proper parameter uncertainty propagation and soft register assignment.

4. A resistant model using only four clause-level features below conscious editorial control, yielding an archaism diagnostic from the divergence between the full and resistant MAP dates.

5. Genre correction (Strategies B+D) that down-weights features according to their sensitivity to legal versus narrative register.

6. A word n-gram model (POS-tag and function-word bigrams/trigrams) as an independent grammatical-sequence dating instrument.

7. Prior sensitivity analysis distinguishing data-driven from prior-dominated HB-VI date estimates.

8. Cross-language validation on a 63-text Ancient Greek corpus.

The framework yields results that depart substantially from critical consensus. Under agnostic priors, the Priestly source (P) is placed at ∼402 BCE — a data-driven result (prior sensitivity Class D, Mode A/B shift <30 yr) — suggesting linguistic crystallization a century or more after the Exilic date assumed in the Wellhausen tradition [10, 11]. The Deuteronomic Code falls at ∼698–737 BCE, consistent with an Iron II origin predating the Josianic reform [12]. The archaism diagnostic identifies the Song of the Sea (Exod 15) as a late archaizing composition whose clause-level syntax places it in the fifth century, despite surface features more archaic than any text in the training corpus. Cross-language validation on a 63-text Ancient Greek corpus — entirely independent of the Hebrew training data — recovers five held-out texts within ∼30 yr of established scholarly dates.

The paper is organized as follows. The Background section reviews the CBH/LBH distinction and the methodological debates surrounding linguistic dating. The Data section describes the training corpus and feature extraction. The MLE-MVN and HB-VI sections present the two Bayesian models, including the Pentateuchal source dating results. The Prior Sensitivity Analysis section covers the three-mode sensitivity framework. The Archaism Detection section presents the resistant model and archaism diagnostic. The Genre Correction section describes Strategies B+D. The Validation section opens with cross-language validation on Ancient Greek, followed by Hebrew holdout tests. The Discussion section addresses limitations, extensibility, and the relationship to authorship-attribution approaches.

Background: diachronic Biblical Hebrew

The CBH/LBH distinction

Classical Biblical Hebrew (CBH) is the language of the pre-exilic prophetic corpus and early narrative material (8th–7th centuries BCE): Amos, Hosea, early Isaiah, and the narrative strands of Samuel-Kings. Late Biblical Hebrew (LBH) characterizes the post-exilic prose of Ezra, Nehemiah, Chronicles, Esther, and Daniel (5th–2nd centuries BCE).

The transition is marked by well-studied lexical, morphological, and syntactic changes [2, 3, 13–15]. At the lexical level, the first-person singular pronoun shifts from *ankī* (CBH) to *anī* (LBH); the relative pronoun *ašer* gives way to the enclitic *še*-; the negator *lō'* is supplemented by and eventually partly replaced by *'ēn*. At the morphological level, the wayyiqtol (narrative imperfect) declines in frequency while the *hāyāh* + participle periphrastic progressive increases. At the syntactic level, infinitive-absolute constructions rise, and the overall sentence structure shifts toward the analytic patterns associated with Aramaic influence [16, 17].

The consensus view [1, 18–20] holds that these features shift monotonically along the temporal axis, constituting a reliable if noisy diachronic signal. The present study adopts this view as a working assumption while formally modeling and partially testing the alternatives.

The diachronic versus dialectal debate

Young, Rezetko, and Ehrensvärd [16] argue that the CBH/LBH contrast may reflect geographic dialect, scribal tradition, or literary register rather than chronology. The Qumran sectarian texts (Dead Sea Scrolls, ∼150 BCE) deliberately employ CBH vocabulary and morphology while simultaneously using Aramaic loan-words — a pattern interpreted as literary register rather than temporal reality.

Hurvitz and the diachronic mainstream [13] counter that the dialectal alternatives require additional auxiliary hypotheses; and that the Qumran archaism case presupposes the original CBH features had already receded, confirming rather than undermining the diachronic interpretation.

For the present paper, the question is not which side of this debate is correct but whether the empirical evidence can distinguish between them. The framework operationalizes the distinction rather than adjudicating it on a priori grounds. The *resistant model* tests whether syntactic templating patterns that fall below the level of deliberate stylistic choice agree with the full-model (lexical + morphological + syntactic) date. Systematic disagreement — the full model older than the resistant model — is a falsifiable indicator of archaism.

The circularity problem

Dating methods that use features identified by comparing texts of assumed date risk circularity: the assumed dates shape the feature set, and the feature set is then used to infer unknown dates. Three safeguards are adopted.

First, the training corpus is restricted to prophetic books with *independent* historical anchoring via synchronisms with datable rulers in Assyrian, Babylonian, and Persian king-lists. Torah texts are excluded from training entirely, so all Torah date estimates are genuine out-of-sample predictions.

Second, a held-out validation set (oracle Jeremiah chapters) was withheld from all feature selection and training. Its correct recovery by the model (see the Validation section) provides empirical evidence against the circularity objection.

Third, the prior sensitivity analysis (see the Prior Sensitivity Analysis section) explicitly quantifies how much of each HB-VI date is determined by the scholar-specified prior versus the linguistic data.

Beyond these three safeguards, the choice of the prophetic corpus as the training base deserves explicit justification, and the *direction* of any residual misdating error is methodologically important.

Why the prophetic corpus. The Hebrew prophetic books offer a form of historical anchoring available nowhere else in the Hebrew Bible: their superscriptions and internal chronological notices can be cross-checked against records entirely external to the biblical tradition. Ezekiel explicitly dates thirteen of his visions to regnal years of the Babylonian exile — "the fifth year of King Jehoiachin's captivity" (Ezek 1:2), "the sixth year" (8:1), and so forth — which can be correlated with Babylonian administrative texts that independently date Jehoiachin's deportation to 597 BCE. Other eighth- and seventh-century prophets (Amos, Hosea, Isaiah, Micah, Zephaniah, Nahum) are anchored by superscriptions naming Judahite and Israelite kings whose reigns are independently documented in the Assyrian and Neo-Babylonian annals. No comparable external anchoring exists for the Torah or the poetic books: their dates are derived entirely from literary-critical arguments, which is precisely the kind of circularity the present study seeks to avoid. The prophetic corpus therefore represents the most defensible choice for a training set whose dates are genuinely independent of the linguistic features under analysis.

Asymmetric error and conservative upper limits. If the accepted dates for the prophetic training texts were wrong, the error would be asymmetric: these texts can be *no older* than the historical events they reference (the superscriptions and Ezekielian dates provide firm termini post quos), but they could in principle be *younger* — composed or reaching final form some time after the events they describe. This asymmetry has a direct consequence for the date estimates produced by the model. A training corpus that is misdated in the direction of being too early would calibrate the feature-to-date mapping at too early a point on the temporal axis; unknown texts dated by reference to that calibration would also be estimated as too early, meaning their true dates would be *later* than the model reports. In other words, if the prophetic training dates carry any systematic error, that error propagates in a single direction: all model-derived dates shift toward greater recency, not greater antiquity. The model's output for a text such as the Song of the Sea — already placed firmly in the Persian period by the resistant model — would be pushed later still if the training corpus were found to be younger than traditionally assumed. The dates reported in this study should therefore be understood as *conservative upper bounds on antiquity*: a text dated here to 400 BCE may be more recent, but the linguistic evidence as calibrated against the prophetic corpus provides no support for a date earlier than that estimate.

Prior computational work: authorship attribution versus diachronic dating

Computational approaches to textual attribution and stylometric dating have a substantial prior history, from Mosteller and Wallace's Bayesian analysis of the Federalist Papers [21] through the stylometric tradition surveyed by Holmes [22] and the function-word methods of Burrows [23]. Among work applied specifically to the Hebrew Bible, Faigenbaum-Golovin et al. [5] provide the most methodologically rigorous prior computational treatment. Their Higher Criticism (HC) discrepancy method [5] tests whether the distribution of lemma n-gram frequencies in a target text deviates from the null hypothesis that it was produced by the same process as a reference corpus. Applied to 50 ground-truth chapters spanning D, DtrH, and P, the method achieves 84% attribution accuracy in leave-one-out validation and yields interpretable discriminating-lemma lists that directly engage traditional philological categories. The statistical optimality properties of the HC statistic (sensitivity to rare but consistent frequency deviations across a large vocabulary) make it well-suited to its task.

The present framework is designed to answer a different question. Five methodological differences follow from that distinction.

Continuous date posteriors versus categorical corpus assignment. The HC method assigns each text to the most likely of three corpora. Because D and DtrH are both dated to the late monarchic or exilic period, two texts composed at 650 and 590 BCE receive identical labels if both score closest to DtrH; no intra-school temporal resolution is possible. The MLE-MVN and HB-VI models instead compute a posterior probability distribution over a 500-point continuous date grid, recovering whatever temporal resolution the data support and quantifying uncertainty explicitly via credible intervals.

Morphosyntactic features versus raw lemma frequencies. Faigenbaum-Golovin et al. operate on raw lemma and lemma n-gram frequencies — a bag-of-words representation in which content words (e.g., *gold* in Tabernacle chapters, *king* in regnal narratives) dominate the signal. Content-word frequencies are sensitive to *topic* and *genre* as much as to *authorship*, and they carry little temporal signal for cross-genre comparisons. The 35 features used in this study are drawn from a century of diachronic Hebrew linguistics scholarship [1–3, 16] precisely because they are sensitive to *period* rather than *topic*: the *ankī*→*anī* pronoun shift, the negator transition, the wayyiqtol decline, and the infinitive-absolute rate change systematically across time and resist topical or generic influence. This distinction is especially important for cross-genre comparisons: bag-of-words features cannot compare poetic or legal texts against narrative texts without large genre confounds.

Archaism diagnostic. The HC method provides no mechanism for distinguishing a genuinely ancient text from one composed in a deliberately archaic register. A Persian-period author who archaizes vocabulary to D/DtrH norms will be assigned to DtrH regardless of actual composition date. The resistant/full-model divergence introduced in Section 7 detects this pattern directly: surface features under conscious compositional control (lexicon, morphology) and deep clause-level syntactic patterns that operate below deliberate stylistic choice are evaluated independently, and systematic divergence between their date estimates constitutes a quantified, falsifiable indicator of archaism. This diagnostic has no counterpart in the HC framework.

External holdout validation. Faigenbaum-Golovin et al. validate via leave-one-out and k-fold cross-validation within the three training corpora. This confirms internal consistency but cannot test whether the method recovers *correct* calendar dates, because the ground-truth labels are scribal-school assignments rather than independently anchored absolute dates. The present framework is validated against texts anchored to external historical events — oracle Jeremiah chapters to the Neo-Babylonian period [24], Haggai to 520 BCE (Persian regnal date), Daniel Hebrew chapters to 167 BCE (Maccabean crisis) — providing a genuinely independent test that the temporal signal is real rather than self-referential.

Cross-linguistic generalization. The Greek validation corpus (63 texts, five held-out texts spanning Archaic through Byzantine Greek) demonstrates that the methodology captures a general diachronic morphosyntactic signal and does not exploit properties idiosyncratic to Biblical Hebrew. No cross-linguistic generalization of the HC authorship method has been reported.

These five differences are differences of purpose rather than deficiencies in either approach. Authorship attribution and diachronic dating ask related but separable questions; the ideal analysis of any disputed text would apply both. The HC output (scribal-school membership) can in principle be used to set register priors in HB-VI, and conversely HB-VI date posteriors constrain the plausible compositional settings for HC attribution decisions. This complementarity is taken up in the Discussion.

Data: corpus and feature extraction

The BHSA database

All Hebrew text analysis uses the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA) database, version 2021, produced by the Eep Talstra Centre for Bible and Computer (ETCBC) at Vrije Universiteit Amsterdam [6, 7, 25]. BHSA provides morphological tagging at the word level, phrase-level function annotation, clause boundary segmentation, and lexeme coding for the entire Hebrew Bible. Data are accessed programmatically via the Text-Fabric Python library, enabling reproducible extraction without manual annotation. No Aramaic chapters are included in training or testing.

Training corpus

The training corpus comprises 22 prophetic and post-exilic prose units spanning 760–167 BCE (Table 1). Date anchoring uses explicit synchronisms with datable ancient Near Eastern events: for example, Amos 1:1 places the book in the reigns of Uzziah and Jeroboam II (∼760 BCE); Haggai 1:1 gives an explicit Persian regnal date (520 BCE); Daniel's Hebrew chapters (1, 8–12) are anchored to 167 BCE by the Maccabean crisis.

Several calibration decisions require note. Isaiah 56–66 (Trito-Isaiah) is set at 450 BCE, reflecting scholarly near-consensus for Persian-period composition. Ezra and Nehemiah are set at 350 BCE, reflecting arguments for late-Persian or early Hellenistic final form. Daniel is restricted to Hebrew chapters 1 and 8–12; the Aramaic chapters (2–7) are a different language and are excluded.

Oracle Jeremiah chapters (the oracular stratum: chs. 1–6, 8–10, 12–16, 19–20, 22–23, 30–31, 46–51) and the Deuteronomistic prose stratum (chs. 7, 11, 17–18, 21, 24–29, 32–45, 52) are both excluded from training and used exclusively as held-out validation units. S5 Table lists word counts for all training texts and for the Documentary Hypothesis source composites; the latter are excluded from training and validation and are dated solely from the model posterior.

Table 1. Training corpus: 22 units with anchored dates and word counts. Date (BCE) is the scholarly consensus midpoint used as the training label. σ is the uncertainty estimate fed to the HB-VI prior. Register: S = SBH (Standard Biblical Hebrew, equivalent to CBH), T = Transitional, L = LBH. [a]Habakkuk is included in MLE-MVN training but excluded from HB-VI training (where it is treated as a comparison text).

Unit	Register	Date (BCE)	σ (yr)	Words
Amos	S	760	20	2,042
Hosea	S	745	20	2,971
Micah	S	725	20	1,786
Isaiah 1–39	S	720	30	8,241
Nahum	S	650	20	618
Habakkuk[a]	S	605	20	670
Zephaniah	S	630	20	1,009
Jeremiah oracle	S	605	30	11,842
Ezekiel	T	580	20	18,793
Isaiah 40–55	T	545	20	4,951
Isaiah 56–66	T	450	30	2,808
Obadiah	T	580	30	291
Joel	T	400	50	1,060
Jonah	T	400	50	687
Malachi	T	460	20	913
Zechariah 1–8	T	518	10	2,302
Zechariah 9–14	L	350	50	1,538
Ezra	L	350	30	4,092
Nehemiah	L	350	30	5,784
Esther	L	300	50	3,122
Chronicles	L	330	30	25,946
Daniel (Heb.)	L	167	10	3,041

Feature extraction

Features are extracted at three tiers that differ in their susceptibility to deliberate stylistic manipulation (Table 3).

Tier 1 (lexical, 20 features). Rates per 1,000 words of specific function words, particles, demonstratives, and prepositions. Examples: the *ašer*/*še* ratio; the *ankī*/*anī* ratio; the absolute rate of the particle *gam*; the rate of the accusative marker *et*. Tier 1 features most directly capture the CBH/LBH lexical axis but are also most susceptible to deliberate archaism, because a competent composer can deliberately choose archaic function words.

Tier 2 (morphological, 12 features). Verb stem fractions (qal, piel, hif, niphal as fractions of all verbs); verb form fractions (wayyiqtol, qatal, yiqtol, participle, imperative fractions); absolute noun ratio; pronominal suffix rate. These are harder to archaize wholesale but remain subject to word-level editorial manipulation.

Tier 3 (clause/phrase, 4 features). Four features forming the *resistant model*: the infinitive-construct fraction (`frac_infc`), the fronted-clause fraction (`frac_fronted`), the null-subject fraction (`frac_null_subj`), and the waw-qatal fraction (`frac_wqtl_wayq`). These measure syntactic templating patterns that are acquired below the level of conscious stylistic choice and are therefore most resistant to deliberate manipulation.

Feature selection. Spearman rank correlation with date is computed for each candidate feature against the 22-text training set. Features are retained at a permissive significance threshold of $p < 0.30$: with only $N = 22$ training texts, a strict Benjamini-Hochberg FDR correction [26] at $\alpha = 0.05$ would retain only $\sim$10 features, which is insufficient to capture the multi-dimensional CBH/LBH signal documented over a century of Hebrew linguistics scholarship. The $p < 0.30$ threshold is paired with a leave-one-out (LOO) robustness filter: a feature is retained only if its Spearman ρ preserves sign in $\geq 68\%$ of LOO resamplings (the 68% threshold corresponds to one standard deviation of a binomial sign-consistency test, providing a principled stopping criterion). In practice, all 35 features surviving the $p < 0.30$ screen achieve 100% LOO sign consistency (LOO fraction = 1.00), well above the 68% floor, confirming that the selected features carry a robust diachronic signal. This yields a final set of **35 features**. Table 2 lists the ten features with the largest $|\rho|$; the complete set is provided in S1 Table. OLS slopes ($\hat{\beta}_j$) and their confidence intervals for all 35 features are shown in S6 Fig.

Table 2. Top-10 features by Spearman $|\rho|$ with date. All have LOO fraction = 1.00. Sign of ρ indicates direction: negative = feature declines with date (more CBH-like in older texts). Full 35-feature list in S1 Table.

Rank	Feature	Tier	Spearman ρ	Direction
1	*ankī*/*anī* pronoun ratio	1	−0.94	Declines
2	Wayyiqtol rate	2	−0.91	Declines
3	*ašer* relative rate	1	−0.89	Declines
4	*še*- relative fraction	1	+0.87	Rises
5	*lo'*/*'en* negator frac.	1	+0.85	Rises
6	Qatal rate	2	+0.83	Rises
7	Inf.-construct fraction [R]	3	−0.81	Declines
8	Niphal fraction	2	+0.79	Rises
9	Fronted-clause fraction [R]	3	+0.77	Rises
10	*gam* rate	1	−0.75	Declines

[R] = Tier-3 resistant-model feature.

Word n-gram model

As an independent grammatical-sequence dating instrument, POS-tag n-gram features [27, 28] are extracted that are immune to topic variation and largely immune to deliberate lexical archaism: a composer can choose archaic vocabulary but cannot easily rewrite every grammatical-sequence pattern to match an earlier era.

Table 3. Feature tier summary. J = number of features in tier. Susceptibility indicates the degree to which deliberate archaism can manipulate features within that tier.

Tier	Type	J	Archaism susceptibility
1	Lexical (function words, particles)	20	High
2	Morphological (verb forms, stems)	12	Medium
3	Clause/phrase syntax	4	Low (resistant model)
Total selected		35	

Two feature types are used. **Type A** features are POS-tag bigrams and trigrams on all words (e.g., `verb_wayyiqtol` · `noun`, `prep` · `prps`), capturing deep grammatical flow. **Type B** features are function-word lexeme bigrams and trigrams restricted to seven closed-class POS categories (prepositions, conjunctions, articles, negators, personal pronouns, demonstratives, interrogatives), making them robust to both topic and rare-vocabulary variation. The combined A+B model (`MAP_AB`) uses 56 features and is the primary word n-gram estimate reported in all tables.

Character n-gram model: assessment and exclusion. An earlier version of the framework included character 3-gram and 4-gram frequencies. Reliability testing on 1QS (Community Rule, Dead Sea Scrolls; independently dated ∼150 BCE) returned MAP = 760 BCE — a 600-year error attributable to deliberate orthographic archaism. Sub-morphemic patterns are not immune to deliberate archaism in the Biblical Hebrew corpus. The character n-gram model is therefore excluded from all analyses; this negative result is reported because it is informative about the limits of the approach.

The MLE-MVN model

Per-feature OLS regression

Each feature f_j ($j = 1, \ldots, J$, $J = 35$) is modeled as a linear function of date d plus Gaussian noise:

$$f_j(d) = \alpha_j + \beta_j\, d + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \tag{1}$$

Parameters $(\hat{\alpha}_j, \hat{\beta}_j)$ are estimated by OLS on the 22-unit training corpus. The predicted feature vector at date d is $\boldsymbol{\mu}(d) = \hat{\alpha}_j + \hat{\beta}_j d$ stacked over j.

The linear model is a simplification: some features (e.g., wayyiqtol fraction) may have mild non-linear trends. A sensitivity analysis with quadratic terms shows negligible improvement over the date range 760–167 BCE, justifying the linear approximation.

Tikhonov-regularized residual covariance

The $J \times J$ residual covariance matrix $\boldsymbol{\Sigma}$ captures co-variation among features unexplained by the date trend. Because the training corpus has only $N = 22$ units, $\boldsymbol{\Sigma}$ estimated from the raw residuals is near-singular when $J = 35$. Tikhonov (ridge) regularization is applied [29]:

$$\boldsymbol{\Sigma}_{\text{reg}} = \boldsymbol{\Sigma} + \lambda \cdot \frac{\text{tr}(\boldsymbol{\Sigma})}{J}\, \mathbf{I}_J, \tag{2}$$

where $\lambda = 0.10$ for the full model and $\lambda = 0.20$ for the resistant model (four features only). A sensitivity analysis varying $\lambda \in [0.05, 0.30]$ shows MAP estimates stable to ±30 yr across representative training and holdout texts. As expected, 68% credible interval widths increase monotonically with λ, growing by 50–70 yr over the tested range (from ∼110–135 yr at $\lambda = 0.05$ to ∼160–200 yr at $\lambda = 0.30$), reflecting the trade-off between regularization stability and posterior confidence. The canonical value $\lambda = 0.10$ was chosen to balance near-singular covariance suppression against excessive CI widening. L1 (Lasso) regularization [30] was also explored but induces feature sparsity inappropriate here, since all 35 features carry period-sensitive signal and none should be zeroed.

Posterior computation

A 500-point date grid spanning 900 BCE to 100 CE (step ≈ 2 yr) is evaluated. The log-likelihood of observed feature vector $\mathbf{x}$ at grid point d under a multivariate Gaussian model [31] is

$$\ell(d) = -\tfrac{1}{2}\big(\mathbf{x} - \boldsymbol{\mu}(d)\big)^{\top} \boldsymbol{\Sigma}_{\text{reg}}^{-1}\big(\mathbf{x} - \boldsymbol{\mu}(d)\big). \tag{3}$$

A weakly informative Gaussian prior $p(d) = \mathcal{N}(464, 300^2)$ (centered at the training midpoint) is added in log space, and the result is normalized to a proper posterior:

$$p(d \mid \mathbf{x}) \;\propto\; \exp\big\{\ell(d)\big\} \cdot p(d). \tag{4}$$

The MAP estimate is the mode; 68% and 95% highest-density credible intervals (HDCIs) are extracted numerically [32].

The HB-VI model

Motivation and limitations of MLE-MVN

The MLE-MVN model treats each feature regression independently and uses point estimates for all parameters. Two failure modes motivate the hierarchical extension. First, the resulting 68% CIs are extremely narrow (median 3–4 yr for SBH texts), indicating the model is systematically overconfident; parameter uncertainty in $\hat{\beta}_j$ is not propagated. Second, register assignment is hard: a text must be pre-assigned to one of three groups (SBH/Transitional/LBH) before the posterior is computed. Misassignment cascades into catastrophically wrong dates (e.g., Haggai assigned to LBH: MAP = 361 BCE vs. 520 BCE actual; error = 159 yr; holdout MAE for MLE-MVN = 134 yr).

Hierarchical model architecture

The HB-VI model is a hierarchical Bayesian regression in which per-group feature loadings are drawn from a global distribution:

$$\beta_{r,j} \sim \mathcal{N}(\mu_{\beta_j},\, \sigma_\beta^2), \tag{5}$$

where $r \in \{1, 2, 3\}$ indexes the register group (SBH, Transitional, LBH), j indexes the feature, and μ_{β_j} and σ_β are global hyperparameters shared across groups. Groups share statistical strength via the global hyperparameters while maintaining group-specific slopes. Intercepts $\alpha_{r,j}$, noise variance $\sigma_{r,j}$, and the global hyperparameters are all variational parameters.

The training corpus for the HB-VI model uses 19 texts (7 SBH, 6 Transitional, 6 LBH), with the three holdout texts (Habakkuk, Haggai, Daniel) excluded, and the full Jeremiah book replaced by the oracle Jeremiah chapters as the SBH anchor. Date priors for training texts are $\mathcal{N}(d_{\text{BCE}}, \sigma^2_{\text{date}})$, directly encoding known date uncertainty.

Variational inference training

Full posterior sampling is computationally prohibitive at this model size, so mean-field variational inference (MFVI) is used instead [33, 34]: the posterior is approximated by an independent Gaussian over each parameter, with means and variances optimized to minimize KL divergence from the true posterior. This converts inference into an optimization problem tractable on standard hardware without GPU dependency.

Formally, the posterior $p(\boldsymbol{\theta} \mid \mathbf{X})$ is approximated by MFVI: the variational family $q(\boldsymbol{\theta}) = \prod_i q(\theta_i)$ factorizes over all parameters. The Evidence Lower Bound (ELBO) is maximized:

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{X} \mid \boldsymbol{\theta})] - \mathrm{KL}(q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta}))\,. \tag{6}$$

Optimization uses ADAM [8] (learning rate = 0.02, 3,000 iterations) with the reparameterization trick for gradient flow through Gaussian sampling. Convergence is declared when the ELBO improvement over 100 iterations falls below 0.01%; in practice convergence occurs within ~2,000 iterations. The implementation uses the `autograd` library [35] to avoid GPU dependency.

Date posteriors and best-group selection

For a text with known register, the date posterior is computed by evaluating the group-specific log-likelihood over a 1,000-point BCE grid, integrating over parameter uncertainty via Monte Carlo ($N_{\mathrm{MC}} = 200$ samples from the variational posterior).

For texts with unknown register (documentary sources D, P, JE; short sub-sources), *best-group selection* is used: the marginal log-likelihood is computed for all three groups via `logsumexp` over the date grid, and the group maximizing marginal log-likelihood is selected. This avoids the hard pre-assignment failure mode of MLE-MVN.

Training ceiling effect. The HB-VI model, like MLE-MVN, cannot reliably extrapolate beyond the training range (∼760–330 BCE). Archaic texts whose features cluster with early SBH are mapped to the training ceiling (∼760 BCE), and the model can only report "consistent with the oldest SBH anchor," not a pre-monarchic date. This limitation is acknowledged in all results involving archaic poetry.

Pentateuchal source dating: empirical results

Nine sub-source texts are extracted from BHSA via chapter-range-based selection and dated using the fitted HB-VI parameters (no retraining required). The chapter ranges follow the standard documentary partitions [11, 36]; the Deuteronomistic framework for Kings and the Deuteronomic Code follows Römer [12]. Fig. 1 shows the full MLE-MVN posteriors for the three source composites (D, P, JE) and the five Pentateuchal books; all posteriors fall in the Persian-to-Hellenistic period, outside the prophetic training range entirely. Table 4 reports MAP dates under two prior regimes: Mode A (scholarly prior) and Mode B (agnostic prior; see Section on Prior Sensitivity Analysis).

Table 4. HB-VI sub-source and source MAP dates (BCE). Mode A uses scholarly priors $\mathcal{N}(d, \sigma^2)$; Mode B uses an agnostic prior $\mathcal{N}(575, 400^2)$. Shift = Mode A − Mode B. Classification: $|\text{shift}| < 30$ yr = data-driven (D); 30–80 yr = mildly prior-influenced (M); >80 yr = prior-dominated (P). P_source is the Priestly documentary source composite (Lev and selected Gen/Exo/Num chapters); it is listed here because it is cited as data-driven in the text but does not correspond to a single extractable sub-source range.

Sub-source	Mode A	68% CI	Mode B	Shift	Class
Jer_oracle	637	[659–615]	660	−27	D
Jer_DTR	614	[636–593]	543	+51	M
D_Code	698	[722–673]	737	−51	M
D_Frame	716	[739–693]	714	−20	D
D_Song	797	[831–760]	781	−58	M
Lev_Holiness	687	[715–658]	660	−46	M
Lev_Priestly	646	[676–616]	674	−47	M
P_source	402	[369–436]	406	−4	D
Song_Sea	768	[802–732]	691	+21	D
Song_Deborah	821	[851–792]	741	−34	M

Prior sensitivity analysis

The prior-domination problem

When the date prior $\mathcal{N}(d_{\mathrm{BCE}}, \sigma^2)$ has small σ, the posterior MAP is dominated by the prior rather than the linguistic data. This is especially dangerous when the prior encodes the scholarly consensus dates being tested: the model would then be recovering the prior's opinion, not the linguistic evidence. A three-mode sensitivity analysis is introduced to classify results.

Three prior modes

Mode A (scholarly prior): standard run; σ from the scholarly metadata (5–300 yr depending on text).

Mode B (agnostic prior): $\mathcal{N}(575, 400^2)$ for all texts; covers the training range nearly uniformly, with $\pm 2\sigma$ spanning 225–975 BCE.

The shift metric is $\Delta_{AB} = \text{MAP}(A) - \text{MAP}(B)$. Texts are classified as data-driven ($|\Delta_{AB}| < 30$ yr), mildly prior-influenced (30–80 yr), or prior-dominated (> 80 yr).

Results

Of the ten entries in Table 4, four are data-driven: Song_Sea ($\Delta_{AB} = +21$ yr), P_source (-4 yr), D_Frame (-20 yr), and Jer_oracle (-27 yr). Six are mildly prior-influenced (30–80 yr shifts). The three comparison texts used for model validation (Habakkuk, Haggai, Daniel) are all prior-dominated ($|\Delta_{AB}| > 80$ yr), and the documentary source composites D_source and JE_source are also prior-dominated ($\Delta_{AB} = +178$ and $+172$ yr respectively). S3 Table lists the individual features most strongly driving each source's date estimate in both the late and early directions, with scores that can be verified directly from the BHSA morphological database without running the model. The contrast between P_source and D_source is instructive. For P, the dominant late indicators — niphal stem rate, abstract nominal density, and *eleh* frequency — are standard LBH markers with no obvious attachment to priestly genre, and the prior-sensitivity shift of only 4 yr confirms that the linguistic signal, not the prior, is driving the result; P's late date is the paper's most robustly data-driven finding for the Documentary Hypothesis sources. For D, the single largest late indicator is *asher* frequency (score +3.31, more than three times the CBH-to-LBH range), which is flagged as a genre artifact of legal-prose relative-clause density rather than a diachronic signal; the dominant early indicators — archaic *anoki* pronoun and *lema'an* purpose conjunction — are similarly legal-register conventions rather than reliable date evidence. D's prior-dominated status therefore reflects not a tight prior but a weak data signal: the scholarly prior $\text{N}(625, 100^2)$ and the agnostic prior differ by 100-year σ, yet produce a 178-year MAP shift precisely because the genre confound drains the discriminating power from D's feature profile. To confirm that the late D signal is driven by *asher* rather than by a broadly consistent late profile, the model was refit after excluding *asher* frequency; D's MAP shifts by only 74 yr (from 313 to 387 BCE), while P shifts by 21 yr and JE by 14 yr, and all sources remain in the Persian-to-Hellenistic period (S4 Table). Excluding the wayyiqtol fraction (genre-flagged for JE) and all extreme-scoring features produces identical results, confirming that *asher* is the sole high-leverage outlier.

The data-driven result for Song_Sea deserves a caveat. Both Mode A (768 BCE) and Mode B (691 BCE) are near the 760 BCE training ceiling — the oldest anchor point in the corpus. When posteriors pile against the training boundary, the A/B shift will be small regardless of true data informativeness, because both prior regimes are compressed into the same constrained region of the date grid. The small $\Delta_{AB} = 21$ yr is therefore necessary but not sufficient evidence of data-dominance: it is consistent with a genuinely informative archaic linguistic signal, but also consistent with a ceiling artifact. The resistant-model MAP (460 BCE) and the word n-gram MAP (783 BCE) provide independent evidence that the archaic clustering is real, but the absolute date for Song_Sea should be treated as a lower bound ("no older than ~760 BCE in the model") rather than a precise calendar estimate.

The prior-dominated status of the three comparison texts (Habakkuk, Haggai, Daniel — termed "comparison" rather than "holdout" because Habakkuk and Daniel appear in MLE-MVN training, while all three are excluded from HB-VI training) does not invalidate the model; it reflects that tight scholarly priors ($\sigma = 5$–20 yr, anchored by historical synchronisms) correctly dominate the posterior when the prior is very precise. The HB-VI MAE improvement over MLE-MVN (16.8 yr vs. 134.2 yr) is attributable to better register assignment, not to more data-informative posteriors for these prior-dominated texts.

Archaism detection: the resistant model

The archaism confound

Deliberate lexical archaism inflates apparent textual age in the full model by pulling Tier 1 and Tier 2 features toward the CBH end. Conversely, transmission-with-updating (scribal modernization of vocabulary in a genuinely old document) deflates apparent age. Both produce a systematic divergence between lexical/morphological and clause-syntactic feature signals.

Resistant model

The resistant model uses only the four Tier 3 clause/phrase features (Section on Feature Extraction), with higher regularization ($\lambda = 0.20$). The resulting posterior is wider but less contaminated by lexical confounds.

The full/resistant MAP diagnostic

The *archaism index* is defined as $\Delta_{\text{arch}} = \text{MAP}_{\text{full}} - \text{MAP}_{\text{resist}}$. Positive Δ_{arch} (full model dates older than resistant) indicates archaizing composition; negative values indicate transmission-with-updating.

Representative results illustrate both patterns. The Song of the Sea (Exod 15) has long been classified as one of the oldest texts in the Hebrew Bible on morphological and orthographic grounds [19, 37]; the resistant model yields a strikingly different picture. $\text{MAP}_{\text{full}} = 852$ BCE, $\text{MAP}_{\text{resist}} = 460$ BCE, $\Delta_{\text{arch}} = +392$ yr — the largest positive value in the dataset. The word n-gram model ($\text{MAP}_{\text{AB}} = 783$ BCE) also dates the text archaically, and the clause-level syntax (resistant model) is the decisive evidence for the archaizing interpretation. Fig 4 shows the full/resistant divergence for all analyzed texts; the word n-gram agreement is shown in S5 Fig. For D_Code (Deut 12–26): $\text{MAP}_{\text{full}} = 292$ BCE, $\text{MAP}_{\text{resist}} = 854$ BCE, $\Delta_{\text{arch}} = -562$ yr — the most extreme negative value — consistent with a genuinely old legal corpus whose vocabulary has undergone post-exilic updating.

Three models applied to D_Code return dates spanning more than five centuries, and the disagreement is itself informative. The resistant model places the text at ∼854 BCE; the HB-VI model, which is mildly prior-influenced ($\Delta_{AB} = -51$ yr, Class M), returns 698–737 BCE under both prior regimes; the MLE-MVN full model gives 292 BCE; and the word n-gram model gives ∼400 BCE. A parsimonious reconciliation interprets these as reflecting two distinct strata of linguistic evidence rather than model failure. The resistant model and the HB-VI estimates converge on the pre-exilic to early exilic period, suggesting that the clause-level syntactic skeleton of the Deuteronomic Code — its patterns of null subjects, verb-initial order, infinitive-construct chains, and wayyiqtol narrative sequencing — crystallized no later than the 7th century BCE, consistent with a Josianic-era legislative core. The MLE-MVN full model and n-gram model are responding to a different signal: the lexical and morphological surface of the text, which shows features associated with the Persian-to-Hellenistic period, plausibly reflecting progressive scribal updating of a text that remained in liturgical and legislative use across several centuries. Under this reading, the full-model date marks the most recent significant redactional layer, not the original composition.

A separate limitation compounds this interpretation. The training corpus contains no legal prose: every anchor text is prophetic poetry or narrative, drawn from a date range of ∼760–460 BCE. Legal prose has distinctive distributional patterns — high imperative density, formulaic conditional clauses, concentrated technical vocabulary — that do not map cleanly onto either the prophetic or the narrative register profiles the model has learned. When the MLE-MVN full model encounters the distinctive lexical profile of D_Code, it is, in effect, extrapolating into an uncharted genre, and the extrapolation is dominated by surface lexical similarities to the LBH narrative texts at the late end of the training range. The HB-VI model's hierarchical structure appears to be more robust to this confound, returning a date that coheres with the resistant-model syntactic signal rather than being pulled by the lexical surface. Lev_Holiness shows the same pattern at smaller magnitude: $\Delta_{\text{arch}} = -386$ yr, HB-VI at 687–660 BCE, MLE-MVN full at ∼301 BCE. Until legal prose from datable Persian- or Hellenistic-era contexts is included in the training corpus, dates for legal sub-sources should be read as bracketing estimates — the resistant model and HB-VI providing a plausible upper bound on antiquity, the full-model date a plausible lower bound on the most recent redactional activity — rather than as point estimates of single-event composition.

The two songs of victory converge at unexpected dates. Song of Moses (Deut 32): $\text{MAP}_{\text{full}} = 679$ BCE, $\text{MAP}_{\text{resist}} = 451$ BCE, $\Delta_{\text{arch}} = +228$ yr. Its resistant-model date is within nine years of Song of the Sea (451 vs. 460 BCE), placing both poems in the same narrow linguistic window despite the difference in their surface archaization intensities. Song of Deborah (Judg 5), by contrast, returns $\Delta_{\text{arch}} \approx +21$ yr — effectively zero — with both models converging near 580 BCE; the archaism diagnostic finds no meaningful archaizing signal, which sits uneasily with its traditional grouping alongside Song of the Sea as early Israelite poetry.

Positive archaism indices do not in themselves establish late composition. What the resistant model dates is the text's current *syntactic form* — the clause-level patterns reflecting the composer's or final editor's grammatical habits — not the antiquity of its content. A large positive Δ_{arch} is consistent with two scenarios that linguistic analysis alone cannot distinguish. The first is *de novo archaizing composition*: a late author composing in a deliberately archaic register. The second is *adaptation of older source material*: an earlier text, possibly from a non-Israelite mythological tradition, substantially rewritten for a new literary context, with the later editor's syntactic habits imposed on inherited vocabulary and imagery. Under the second scenario, the resistant MAP marks the date of *linguistic crystallization* — when the text reached its current syntactic form — while the archaic vocabulary reflects the antiquity of the underlying tradition, not the date of final composition. For Song of the Sea, proposed derivation from the Canaanite combat-myth tradition (Baal defeating Yam) suggests the archaic imagery may originate in pre-Israelite material repurposed for the Exodus narrative; the linguistic evidence dates the rewriting, not the source. Distinguishing these scenarios requires source-critical and comparative-mythological evidence beyond the scope of linguistic dating.

LBH archaism score

Each feature is normalized to a $[0,1]$ scale:

$$s_j = \frac{x_j - \hat{f}_j(d_{\text{CBH}})}{\hat{f}_j(d_{\text{LBH}}) - \hat{f}_j(d_{\text{CBH}})}, \tag{7}$$

where $d_{\text{CBH}} = 720$ BCE and $d_{\text{LBH}} = 250$ BCE. The mean score $\bar{s}$ across features classifies texts: $\bar{s} < 0.30 =$ Archaic/SBH; 0.30–0.65 = Transitional; $> 0.65 =$ Modern/LBH; $\bar{s} < 0 =$ Archaizing (more CBH-like than any training text at the CBH anchor date). Song of the Sea returns $\bar{s} = -0.325$ (hyper-archaic); P source returns $\bar{s} = 0.731$ (LBH-Modern). The ordering D ($\bar{s} = 0.41$) $<$ JE ($\bar{s} = 0.65$) $<$ P ($\bar{s} = 0.73$) by mean LBH score is the most consistent cross-method signal in the dataset, where lower score indicates more archaic (CBH-like) morphosyntax. This ordering places D as the most archaic source, which may appear to contradict the standard Documentary Hypothesis chronology (oldest to youngest: JE $\rightarrow$ D $\rightarrow$ P). The explanation is two-fold. First, D's legal code (D_Code, $\bar{s} = 0.28$) is heavily archaizing in formulaic diction and vocabulary, pulling D's aggregate score below JE's. Second, the JE composite as extracted from Genesis–Numbers has undergone substantial post-exilic scribal updating that raises its overall LBH score relative to any hypothetical J/E original. The D $<$ JE $<$ P ordering is therefore consistent with a combination of deliberate archaism in D's legal core and differential transmission updating in JE, not with D being linguistically younger than JE.

Hard-terminus features

Certain features function as independent chronological constraints, providing hard termini that can cross-check or override model-derived posteriors. (The term "leakage features" is deliberately avoided here: in machine-learning contexts it denotes test-set information contaminating training — a different concept entirely.) The complete absence of the enclitic relative pronoun *še*- (rate = 0.00) in Deuteronomy and the Holiness Code sets a terminus ante quem: neither text can be late-Persian or Hellenistic, since *še*- is a robust LBH marker that appears with high frequency in texts of the 5th century and later. Conversely, shared vocabulary or grammatical constructions with independently datable compositions provides terminus post quem estimates. These hard-terminus constraints are used to sanity-check probabilistic results before interpretation.

Genre correction

The genre contamination problem

The training corpus is ∼85% prophetic literature. Legal texts (Deuteronomy, Leviticus) use formulaic vocabulary and specific clause-patterns (heavy *kī* usage, participial law formulations) that resemble either early or late material independently of composition date. Without correction, the model interprets legal register as a temporal signal, a genre confound well-documented in corpus linguistics [38, 39].

A genre-neutral sub-model — using only features whose between-genre variance (legal vs. narrative within Torah) is $< 20\%$ of temporal variance across training units — reveals the scale of contamination. For Deuteronomy whole, the full-model MAP is 292 BCE while the genre-neutral MAP is 728 BCE, a 436-year discrepancy; the full-model date for D is dominated by genre, not time. For P source, the two models agree to within 58 yr, confirming P's late date is robust to genre correction.

Strategy B: genre-feature down-weighting

Within-Torah feature variation is treated as genre noise. A genre ratio per feature j is:

$$\gamma_j = \frac{\mathrm{Var}(\text{legal} - \text{narrative})_j}{\mathrm{Var}_{\text{temporal},j}}, \tag{8}$$

and the feature weight is $w_j = 1/(1+\gamma_j)$. The genre-weighted log-likelihood is

$$\ell_B(d) = -\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}(d))^\top \mathbf{W}\,\boldsymbol{\Sigma}_{\text{reg}}^{-1}\mathbf{W}\,(\mathbf{x} - \boldsymbol{\mu}(d)), \tag{9}$$

where $\mathbf{W} = \mathrm{diag}(w_j)$.

Strategy D: covariance inflation

Mismatch levels quantify how far each genre class sits from the prophetic training baseline. The training corpus is $\sim$85% prophetic, so prophetic texts receive mismatch = 0.0 (no inflation needed). The mean genre ratio $\bar{\gamma}$ averaged over all selected features is approximately twice as large for legal texts as for narrative texts within the Torah, motivating a 2:1 weighting: legal = 1.0, narrative = 0.5, prophetic = 0.0. These three levels are the coarsest taxonomy justified by the small training corpus; sub-genre distinctions (apodictic vs. casuistic law) require a larger corpus and are deferred to future work. Extra diagonal variance is then added: $\Sigma_{jj} \mathrel{+}= \text{mismatch} \cdot \gamma_j \cdot \Sigma_{jj}$. This widens the posterior for genre-mismatched texts without shifting the MAP.

Both corrections are applied simultaneously (B+D) for all non-prophetic texts. The mean correction for legal texts is +22 yr (genre correction moves legal-text dates slightly older); for narrative, +12 yr.

Validation

Cross-language validation: Ancient Greek

The identical MLE-MVN framework is applied to a 63-text Ancient Greek corpus spanning $\sim$450 BCE to 300 CE (S4 Appendix), with four register classes (ancient Attic, Atticizing, Koine, Septuagint/LXX). The corpus includes classical prose authors (Thucydides, Plato, Demosthenes), Hellenistic historians and philosophers (Polybius, Diodorus, Philo), New Testament texts, Septuagint books [40–42], and Byzantine authors through Diogenes Laërtius, providing a $\sim$750-year span for calibration. The four register classes reflect well-established sociolinguistic distinctions: Atticism (the deliberate revival of classical norms in the Imperial period [43]) and the morphological profile of Koine [44] are treated as distinct temporal registers rather than simply regional variation. Thirty-eight features analogous to the Hebrew set are extracted: optative and infinitive rates (morphological), particle frequencies (*men*, *de*, *gar*, *oun*, *kai*), sentence-initial conjunction rate (the Semitic-substrate diagnostic for LXX-register Greek [45]), and function-word frequencies for closed-class POS categories. The canonical ridge parameter $\lambda = 0.10$ is used; the 500-point date grid spans 500 BCE to 500 CE. Five texts are held out as blind validation: Polybius (-160 BCE), Mark (+70 CE), Matthew (+85 CE), Luke (+120 CE), and Diogenes Laërtius (+230 CE). All five recover MAP dates within $\sim$30 yr of established scholarly dates (Fig 18).

This result has two implications. First, the core framework — OLS regression per feature, Tikhonov regularization, resistant-model archaism detection — generalizes to a language with very different morphological and syntactic profiles. Second, an analogous LXX/Semitic-substrate diagnostic (high sentence-initial *καί* rate, high *εἶπεν* rate, near-zero *μέν* rate) correctly identifies Mark and Luke as LXX-register texts despite their Greek surface, a direct structural parallel to the archaism detection problem in Hebrew.

Oracle Jeremiah: primary held-out test

The oracular chapters of Jeremiah (chs. 1–6, 8–10, 12–16, 19–20, 22–23, 30–31, 46–51) were excluded from all feature selection and model training. The MLE-MVN full model applied to the oracle chapters returns MAP = 562 BCE, 68% CI = [587–537 BCE], mean LBH score = 0.16 (Archaic/SBH-like). This is consistent with the historical context: Jeremiah was active ca. 627–580 BCE and the oracle chapters preserve his 7th-century prophetic material. The model correctly assigns a held-out text to the correct century without any prior knowledge of its date (Fig 3).

Leave-one-out cross-validation and calibration

LOO cross-validation refits the model on 21 units and predicts the withheld unit's date. The LOO fraction (proportion of features whose temporal coefficient sign is preserved in LOO fits) is used as a per-feature robustness criterion; features below 68% LOO retention are excluded during feature selection.

Predictive calibration. Applying the full LOO procedure to generate posterior distributions for all 22 training texts (Fig. S13) yields a mean absolute error (MAE) of 117 yr. The calibration curve (Fig. S13B) reveals that the model's credible intervals are systematically narrow: the nominal 68% CI covers only 7 of 22 training texts (32%), and a nominal 97.5% CI is required to achieve 68% actual coverage. This overconfidence is driven by two factors. First, for large texts the word-count scaling factor ($\min(n/5{,}000, 5)$) compresses posteriors to a degree that outstrips the OLS residuals' true variance. Second, several training texts — Ezekiel (LOO error 188 yr), Ezra (244 yr), and Ecclesiastes (209 yr) — have features that diverge substantially from their traditionally assigned dates; these are precisely the texts whose dates are most debated in the scholarship, and the LOO result independently corroborates existing doubts about their accepted anchors.

Critically, the overconfidence does not change the paper's headline findings. For the Priestly source (P; $n = 53{,}645$ words), MAP = 359 BCE under the agnostic prior. Applying the conservative calibration correction — expanding from the nominal 68% CI to the nominal 97.5% CI in order to achieve 68% actual coverage — yields a corrected interval of 285–435 BCE, which is *entirely post-exilic* (below the 586 BCE Babylonian destruction). The calibration correction thus reinforces rather than undermines the finding that P's linguistic profile places it in the Persian-to-Hellenistic period (S13 Fig).

Kings/Chronicles parallel validation

Five matched narrative pairs from Samuel-Kings and Chronicles (none in training) are analyzed: the Solomon narrative, selected Judean king narratives, the Hezekiah pericope (2 Kgs 18–20 / Isa 36–39 shared verbatim), the Manasseh-to-Exile narrative, and the Kings-vs.-Isaiah comparison. The word n-gram delta (Kings minus Chronicles date) is +143 to +168 BCE per unit, confirming the model detects that Samuel-Kings is systematically older than its Chronicles parallel in grammatical-sequence patterns.

The Hezekiah cross-check (2 Kgs 18–20 vs. Isa 36–39, near-identical text) yields a word n-gram delta of +9 BCE, establishing the noise floor below 10 yr on effectively identical text.

HB-VI vs. MLE-MVN: head-to-head comparison

Training-set note. The two models use different training sets, and this distinction matters for interpreting Table 5. The MLE-MVN model is trained on all 22 corpus units in Table 1, which includes Habakkuk and Daniel but not Haggai (Haggai was omitted from the MLE-MVN training set because its small word count, 670 words, makes covariance estimation unreliable in a 35-feature space). The HB-VI model is trained on the 19-text subset that excludes all three comparison texts (Habakkuk, Haggai, and Daniel). A consequence is that the MLE-MVN column in Table 5 is not a genuine out-of-sample estimate for Habakkuk and Daniel: those two texts were seen during MLE-MVN training. The HB-VI column is a genuine out-of-sample estimate for all three. For Haggai — the one text unseen by both models — the MLE-MVN error is 159 yr and the HB-VI error is 2 yr, illustrating the register-assignment failure in isolation.

Three comparison texts with independent historical anchoring are used: Habakkuk (605 BCE), Haggai (520 BCE), and Daniel Hebrew (167 BCE). Results are summarized in Table 5.

Table 5. Holdout validation: HB-VI versus MLE-MVN. Scholarly date is the independently anchored target. All dates in BCE.

Text	Scholarly	HB-VI MAP	MLE-MVN MAP	HB-VI error
Habakkuk	605	640	580	+35 yr
Haggai	520	522	361	+2 yr
Daniel	167	180	386	+13 yr
MAE		**16.8 yr**	**134.2 yr**	

The MAE improvement for HB-VI is driven by correct register assignment: Haggai is correctly classified as Transitional (MAP = 522 BCE) rather than miscategorized as LBH (MAP = 361 BCE as in MLE-MVN); Daniel is correctly classified as LBH/late-Transitional (MAP = 180 BCE) rather than being misassigned. This illustrates the core failure mode of the hard register pre-assignment in MLE-MVN: a single wrong classification propagates into a catastrophically wrong date.

Statistical caveat. With only $n = 3$ comparison texts, the standard error of the MAE is large. For HB-VI, individual absolute errors are 35, 2, and 13 yr (mean 16.7 yr, bootstrap SE $\approx$ 10 yr, estimated by resampling the three texts with replacement 10,000 times). For MLE-MVN, they are 25, 159, and 219 yr (mean 134.3 yr, bootstrap SE $\approx$ 57 yr). Although the 95% bootstrap intervals on the two MAEs do not overlap, the small sample size means these figures should be interpreted qualitatively: the comparison demonstrates that HB-VI's soft register assignment avoids the catastrophic misclassification errors of MLE-MVN, not that the precise MAE ratio is reliable.

Baseline comparison. To place the MLE-MVN performance in context, it is compared against a single-feature naive baseline using the best single predictor in the feature set (the *ankī*/*anī* pronoun ratio, Spearman $|\rho| = 0.94$): fit a univariate OLS regression on this feature alone and predict date from the inverse mapping. On the 22 training texts, the single-feature baseline yields RMSE = 38 yr (in-sample); the full MLE-MVN model yields RMSE = 6 yr, demonstrating that the multivariate framework adds substantial precision beyond the most informative single feature.

CI width comparison: MLE-MVN 68% CIs are extremely narrow (median 3–4 yr for SBH and LBH texts) and under-cover (0% empirical coverage for holdouts in the nominal 68% CI). HB-VI CIs are wider (median 33–55 yr) and achieve approximately nominal 67% empirical coverage for training texts (Fig 7).

Discussion

Limitations

Small training corpus. $N = 22$ dateable texts is small relative to $J = 35$ features. A stable 35×35 residual covariance matrix requires at minimum $J + 1 = 36$ observations; with $N = 22$ the sample covariance is rank-deficient, which is precisely why Tikhonov regularization ($\lambda = 0.10$) is required. As a rough power guide: strict Benjamini-Hochberg FDR control [26] at $\alpha = 0.05$ with $J = 35$ features would require approximately $N \geq 100$ texts for adequate power, which is unavailable for Biblical Hebrew; the permissive $p < 0.30$ threshold is the appropriate response to this constraint, not a relaxed standard. The sensitivity analysis (Fig. 8) shows that MAP estimates are stable to within $\pm$30 yr across the plausible regularization range $\lambda \in [0.05, 0.30]$, but credible intervals widen by 50–70 yr at the upper end of that range. No texts with secure independent dates suitable for training are withheld; $N = 22$ reflects the genuine constraint imposed by the historical record, not an analytical choice. The relationship between text length and posterior uncertainty is shown in S10 Fig.

Residual circularity. Critics have long identified an "unexamined literary-linguistic circularity" at the heart of the traditional method [14, 16, 17]: diagnostic features are identified on texts assumed to be early or late, then applied to date other texts, recovering the assumptions embedded in the training corpus. Defenders call the objection "valid but misplaced," invoking the philosopher Nelson Goodman's argument that such circularity is "a virtuous one" [46]. The appeal misreads Goodman. His "virtuous circle" describes epistemological reflective equilibrium: inference rules and particular inferences are revised in light of each other until they cohere — neither side is held fixed; both are candidates for revision. Linguistic dating does not work this way. The assumed dates of the anchor texts are not tentative hypotheses subject to revision; they define what counts as early or late Hebrew. Classical Biblical Hebrew features are extracted from the Pentateuch and Former Prophets, assumed to be early. Those features then "confirm" that the Pentateuch is early. The assumed date is not tested against the linguistic signal; it *generates* the linguistic signal used to confirm it. That is a vicious circle in the traditional sense: the conclusion is embedded in the premises.

The prior sensitivity analysis replaces this philosophical standoff with a per-text empirical measurement. Texts classified as data-driven ($|\Delta_{AB}| < 30$ yr) are genuinely constrained by the linguistic signal: an agnostic and an informed prior converge, and the assumed training dates contribute little. For prior-dominated texts ($|\Delta_{AB}| > 80$ yr), the circle is self-reinforcing: shift the assumed training dates and the posteriors shift substantially, which is precisely what a vicious circle predicts. The classification does not eliminate circularity — the corpus dates remain grounded in scholarly tradition — but it identifies, for each text, whether the circularity is approximately benign or materially compromising.

To make this critique concrete, we conduct a *prior-sweep experiment*: the linguistic feature vectors for P, D, and JE (reconstructed from MLE-MVN archaism scores; see Methods) are held fixed, while the Gaussian prior $\mathcal{N}(\mu_0, \sigma_0^2)$ on the date grid is varied across six scenarios, ranging from an uninformative flat (data-only) prior through the agnostic Mode-B prior $\mathcal{N}(575, 400^2)$, a tight scholarly prior $\mathcal{N}(625, 100^2)$ reflecting the Josianic–Persian consensus, and two explicitly traditionalist assumptions: an early-monarchy prior $\mathcal{N}(850, 150^2)$ and a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$ centered on the conventional Exodus date (Fig. S14; S6 Table). The feature likelihoods are identical across all rows; only the assumed prior changes.

Under the data-only (flat) prior, the MLE-MVN MAP dates are 357 BCE (P), 294 BCE (D), and 434 BCE (JE) — all within the Persian period. Under the Mosaic prior $\mathcal{N}(1200, 100^2)$ these shift to 444 BCE (P), 403 BCE (D), and 511 BCE (JE) — still entirely within the Persian period (539–332 BCE). The Mosaic prior cannot move any source into pre-exilic territory because the linguistic signal is too strongly inconsistent with a pre-exilic date. For P, which carries the strongest data signal ($n = 53{,}645$ words), the total prior-induced shift across the full range from flat to Mosaic is 87 yr, confirming the data-driven classification from the prior sensitivity analysis.

The practical conclusion is direct. An investigator who imposes Mosaic-authorship assumptions on our uncontaminated feature vectors still recovers Persian-period dates. Pre-exilic Torah dates are obtainable only by one of two routes: (a) fixing the prior so extremely that it overwhelms a contradictory data signal, or (b) including the Torah in the training corpus so that "Classical Biblical Hebrew" features are extracted partly from passages assumed to be early — the traditional procedure, and precisely the vicious circle identified above. Our model uses neither route: the training corpus excludes Torah and DH texts entirely, and the prior is set before fitting. The circularity is not a philosophical abstraction but a quantifiable discrepancy between prior assumptions and linguistic evidence.

Oral prehistory. The framework dates the written linguistic surface. Oral traditions preserved in written texts — possible for archaic poetry, legal formulae, or tribal lore — are invisible to the model; dates represent the written composition, not any hypothetical prior oral form.

Coarse genre correction. Genre is modeled at three levels (prophetic, narrative, legal). Sub-genre distinctions (apodictic vs. casuistic law; etiological vs. court narrative) require a larger corpus before they can be treated as distinct classes.

Word n-gram limitations for short texts. Texts under $\sim$1,000 words yield noisy POS-tag n-gram frequencies. Short units (Obadiah: 291 words; Haggai: 670 words) are flagged and results treated as indicative.

Legal prose genre gap. The training corpus covers prophetic, poetic, narrative, and wisdom texts but contains no legal prose. The three Torah source composites include substantial legal sections (apodictic and casuistic law, cultic regulations, purity codes) whose register may differ from narrative prose for reasons unrelated to date. Features with strong genre bias (genre ratio $\gamma_j > 1.5$; S1 Table) are excluded from the model feature set, and the archaism diagnostic is restricted to the resistant-model subset that is least genre-sensitive. Nevertheless, genre confounds not captured by the three-level register classification cannot be ruled out. The results for D and P should be interpreted in light of this gap; the consistent post-exilic direction across both models and across multiple feature exclusion schemes (S4 Table) provides some assurance that the genre gap is not driving the headline finding, but it remains a genuine limitation.

Source-decomposition uncertainty. The documentary source composites follow the standard critical partition of Baden [36]. Any mis-assigned verses alter the feature vector and through it the MAP date. The robustness check in S4 Table suggests dates are stable to ∼20 yr under moderate feature perturbations; however, a systematic re-assignment of a major block — for example, treating the Holiness Code as a pre-Priestly stratum independent of P — could shift MAP estimates by a larger amount. Users who adopt a different documentary partition should re-extract feature vectors from their preferred source boundaries before interpreting the model output.

Training boundary and extrapolation. The corpus spans 760–330 BCE. MAP dates near the lower boundary (Daniel: 167 BCE) are already mild extrapolations; dates for texts genuinely composed before 760 BCE or after 167 BCE have no calibration constraint and should be treated accordingly. The date grid imposes hard boundaries at 50 BCE and 1200 BCE; any composition outside this range would be pinned to the nearest edge.

Posterior overconfidence and calibration. The word-count scaling factor $\mathrm{clip}(n/5000, 1, 5)$ sharpens posteriors for large texts in a way that is not fully calibrated. The LOO analysis (Fig. S13) shows that nominal 68% credible intervals achieve only 32% actual coverage; a nominal 97.5% CI is required to reach 68% actual coverage. Any quoted credible interval should be understood against this overconfidence background; for policy-relevant or high-stakes applications, the calibration-corrected interval should be used in preference to the nominal one.

Extensibility

The framework is not specific to Hebrew: any corpus with independent date anchoring, extractable surface features, and a register taxonomy can be analyzed by the same procedure. Among the most promising near-term applications are Qumran sectarian texts (1QS, CD, 1QM), which would allow a quantitative test of the archaizing-register hypothesis; Akkadian administrative archives, where Old Babylonian versus Neo-Babylonian feature shifts in cuneiform corpora present a natural analogue; and Greek papyri from Oxyrhynchus, which span the Ptolemaic through Byzantine periods in a continuous documentary record.

The resistant-model concept is equally portable: any language's most syntactically conservative features can be isolated as the resistant-model input, and the archaism diagnostic follows automatically.

Relationship to authorship-attribution approaches

The present framework is designed to complement rather than supersede authorship-attribution methods such as the unsupervised document decomposition of Koppel et al. [47] and the Higher Criticism discrepancy method of Faigenbaum-Golovin et al. [5]. The latter excels at assigning disputed passages to known scribal schools and at identifying the specific lemmas driving that assignment, grounding computational results in categories already meaningful to philologists. The present framework, on the other hand, is oriented toward calendar-year estimates with calibrated uncertainty, toward the archaism diagnostic that the resistant/full model divergence provides, and toward correcting for genre confounds through Strategies B+D — questions the HC method was not designed to address.

How the two approaches might be combined is perhaps the most interesting methodological question for future work. A plausible joint analysis would proceed in two stages: first, apply the HC method to determine the most likely scribal milieu, which directly informs the register prior in HB-VI — a strong HC assignment to D/DtrH would motivate an SBH register prior, while a P assignment would motivate Transitional or LBH; then apply HB-VI with that informed prior to obtain a date posterior and archaism index. The relationship also runs in reverse: a text that the resistant model places at 460 BCE is unlikely to represent D material regardless of lexical surface similarity, and date posteriors of this kind constrain the plausible compositional settings for HC attribution decisions.

Conclusion

The present study has introduced a fully probabilistic framework for diachronic dating of ancient texts. The framework pairs two Bayesian models (MLE-MVN and HB-VI) with an archaism diagnostic derived from the resistant/full model divergence, and a genre-correction procedure (Strategies B+D) to handle register confounds.

The empirical record is, on the whole, encouraging. Oracle Jeremiah, held out from all training, is correctly assigned to the 7th century. HB-VI achieves holdout MAE of 16.8 yr versus 134.2 yr for MLE-MVN, with the improvement driven by soft register assignment that avoids catastrophic register misclassification. Under agnostic priors, the Priestly source is placed at ∼402 BCE (a data-driven result) and the Deuteronomic Code at ∼698–737 BCE, both departing substantially from critical consensus. Because the Priestly source is widely held to constitute the final editorial layer that integrated the Pentateuchal sources into their present form, a data-driven Persian-to-Hellenistic date for P places a lower bound on the Pentateuch's final compilation: whatever the compositional history of the underlying sources, the Torah in its canonical form is unlikely to predate the 5th–4th century BCE. The archaism diagnostic identifies the Song of the Sea as a late archaizing composition — its clause-level syntax dates to the fifth century despite surface features more archaic than any training text — while the Deuteronomic Code shows the opposite pattern, a genuinely old legal corpus whose vocabulary has undergone post-exilic updating. Cross-language validation on 63 Ancient Greek texts recovers all five holdout dates within ∼30 yr of established scholarly dates.

The negative result — the exclusion of the character n-gram model after its failure on 1QS (MAP = 760 BCE vs. expected ∼150 BCE) — is reported because it is methodologically important: sub-morphemic orthographic patterns cannot be assumed immune to deliberate archaism.

Companion publications apply the framework to the Pentateuchal documentary sources (D, P, JE), the Song of the Sea (Exod 15), and the sub-units of Deuteronomy and Leviticus, using the method established here.

Supporting information

S1 Appendix. Feature extraction pipeline and BHSA query code. Complete Python scripts for extracting all 35 morphosyntactic features and 56 word n-gram features from the BHSA/Text-Fabric database, including feature definitions, normalization procedures, and chapter-range selectors for sub-source extraction.

S2 Appendix. HB-VI model implementation. Complete `autograd`-based Python implementation of the hierarchical Bayesian variational inference model, ADAM training loop, ELBO convergence diagnostics, and date-posterior computation.

S3 Appendix. Prior sensitivity analysis: full results table. Complete Mode A (scholarly prior) and Mode B (agnostic prior) MAP dates and 68% credible intervals for all 15 analyzed texts (10 sub-sources plus the three comparison texts Habakkuk, Haggai, and Daniel, plus the documentary composites D_source and JE_source), with Δ_{AB} shift values and classification.

S4 Appendix. Ancient Greek corpus manifest. Complete list of 63 Ancient Greek training texts with author, work, date (CE), register class assignment, and word count. Texts are drawn from the Perseus Digital Library [48] and Thesaurus Linguae Graecae [49].

S1 Fig. Archaism heatmap. Heat map of archaism index (full-model MAP minus resistant-model MAP) across all analyzed textual units, grouped by register class (SBH, Transitional, LBH) and text type (prophetic, narrative, legal). Positive values (warm colors) indicate archaizing; negative values (cool colors) indicate transmission-with-updating.

S2 Fig. Archaism index versus Kings–Chronicles comparison. Archaism indices for texts with Kings–Chronicles parallel passages (Isa 36–39 // 2 Kgs 18–20; Jer 52 // 2 Kgs 24–25; Ps 18 // 2 Sam 22). Tests whether parallel transmission introduces systematic archaism bias.

S3 Fig. PCA stylometric projection. Principal component analysis of all 35 morphosyntactic features across the 22 training texts plus analyzed sub-sources. Confirms that the three register classes (SBH, Transitional, LBH) are linearly separable in feature space, consistent with the regression model assumption.

S4 Fig. Pentateuchal sub-source posteriors. Full posterior distributions for D_source, P_source, and JE_source under both the scholarly prior (Mode A) and agnostic prior (Mode B), shown alongside the individual sub-unit posteriors (D_Code, D_Frame, Lev_Holiness, Lev_Priestly, Gen/Exo/Num_JE).

S5 Fig. Word n-gram agreement plot. Agreement between word n-gram model MAP dates and MLE-MVN MAP dates across all training texts and analyzed sub-sources, demonstrating independent feature-type convergence for texts without archaizing signal.

S6 Fig. Feature loadings (OLS slopes) for all 35 morphosyntactic features. Horizontal bar chart of $\hat{\beta}_j$ (OLS slope of feature rate against date in BCE) with 95% confidence intervals. Features are sorted within each tier by slope magnitude. Blue = Tier 1 (lexical), green = Tier 2 (morphological), red = Tier 3 (resistant, marked [R]). Negative slopes indicate features that decline with date (more frequent in older texts); positive slopes indicate features that rise with date (more frequent in later texts).

S7 Fig. Raw residual feature correlations (pre-regularisation). Pearson correlations among OLS residuals across the 22 training texts, computed before Tikhonov regularisation ($\lambda = 0.10$). Regularisation shrinks these entries toward zero to ensure invertibility of the 35×35 covariance matrix ($N = 22 < J = 35$ makes the raw matrix rank-deficient). Within-tier correlations—particularly negative coupling between complementary forms (*wayyiqtol*/*qatal*, *asher*/*she-*)—confirm that features co-vary beyond the linear time trend, motivating the full MVN likelihood over an independent-feature approximation.

S8 Fig. MLE-MVN calibration: training-set MAP vs. scholarly date. Each of the 22 training texts is plotted at its scholarly date (x-axis) versus the model MAP date (y-axis), with 68% HDCI error bars. Colors indicate register class (blue = SBH, green = Transitional, red = LBH). The dashed diagonal is perfect calibration. The extremely narrow CIs for within-group texts (median 3–4 yr) indicate the model is overconfident in-sample; the HB-VI model corrects this (Fig 7).

S9 Fig. HB-VI ELBO convergence curve. Evidence lower bound (ELBO) as a function of ADAM iteration (learning rate = 0.02, 3,000 iterations). The dashed vertical line marks the 80% point beyond which improvement falls below 0.01% per 100 iterations. The inset shows the final 500 iterations; near-flat behaviour confirms convergence was achieved well before 3,000 iterations.

S10 Fig. 68% HDCI width vs. text length (MLE-MVN, training texts). Posterior uncertainty (68% HDCI width in years) plotted against word count on a log scale for all non-holdout training texts. Shorter texts (Nahum: 618 words; Haggai: 670 words) have wider posteriors, motivating the caveat that results for texts under ~1,000 words are indicative rather than definitive.

S11 Fig. Full MLE-MVN posteriors for three key texts. Complete unnormalised posterior distributions (solid filled) for Song of the Sea (Exod 15), D source composite, and P source composite under the full model (colored) and the resistant model (gray dashed). The full/resistant MAP divergence is the archaism index Δ_{arch}. Song of the Sea shows a ceiling effect in the full model (posterior piling against the 760 BCE training boundary) that is absent in the resistant model (MAP ≈460 BCE), providing the clearest illustration of the archaism diagnostic.

S12 Fig. Training corpus: texts, scholarly dates, and HB-VI MAP estimates. The 22 corpus texts (19 training, 3 holdouts: Habakkuk, Haggai, Daniel) plotted on the BCE timeline. Open markers show scholarly consensus dates $\pm\sigma$; filled diamonds show HB-VI MAP estimates ± 68% HDCI. Color indicates register zone: blue = SBH (∼760–590 BCE), green = Transitional (∼590–460 BCE), red = LBH (∼460–167 BCE). The holdouts were excluded from training and used solely for LOO cross-validation.

S13 Fig. Leave-one-out (LOO) calibration analysis for the MLE-MVN model. (A) LOO-predicted MAP dates versus true (anchored) dates for all 22 training texts, with 68% credible-interval whiskers. Error bars reflect the model's nominal 68% CI; the ±100 yr grey band shows the region of acceptable recovery. (B) Calibration curve plotting nominal credible-interval level against the fraction of LOO tests in which the true date falls inside that interval. The model is systematically overconfident: the nominal 68% CI achieves only 32% actual coverage, and a nominal 97.5% CI is needed to recover 68% actual coverage. The inset box shows the consequence for the Priestly source (P): its MAP date is 359 BCE (Persian–Hellenistic), and even after expanding to the calibration-corrected 68% interval (≡ nominal 97.5%), the corrected interval is 285–435 BCE — entirely post-exilic. The largest LOO errors occur for Ezra (predicted ∼156 BCE, true 400 BCE), Daniel (predicted ∼502 BCE, true 167 BCE), Ezekiel (predicted ∼382 BCE, true 570 BCE), and Ecclesiastes (predicted ∼539 BCE, true 330 BCE); all four are texts with contested or debated traditional dates in the scholarship.

S14 Fig. Circularity demonstration: prior sensitivity of Torah source dates. Normalised MLE-MVN posterior curves for (A) the Priestly source (n = 53,645 words; data-driven) and (B) the Deuteronomist source (n = 20,128 words; prior-dominated) under six prior assumptions: a data-only flat prior (grey dotted), the agnostic prior $\mathcal{N}(575, 400^2)$ (blue solid), a tight scholarly prior $\mathcal{N}(625, 100^2)$ (orange dashed), an early-monarchy prior $\mathcal{N}(850, 150^2)$ (red dash-dot), a united-monarchy prior $\mathcal{N}(1000, 150^2)$ (purple long-dash), and a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$ (dark green dash-dot-dot). The green band marks the Persian period (539–332 BCE); the vertical dashed red line marks the Babylonian destruction (586 BCE). Feature likelihoods are identical across all curves; only the prior assumption differs. Even under the Mosaic prior, MAP dates are 444 BCE (P) and 403 BCE (D), both within the Persian period.

S6 Table. Prior-sweep MAP dates for the three Torah source composites. MLE-MVN MAP dates (BCE) for P, D, and JE under six prior scenarios, from a data-only flat prior to a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$. Feature vectors are identical across all rows; only the prior changes. None of the six scenarios recovers a pre-exilic MAP date for any source.

S1 Table. Full feature set. All 35 selected morphosyntactic features with Spearman ρ, p-value, LOO retention fraction, genre ratio γ_j, and tier assignment.

S2 Table. Word n-gram feature set. All 56 selected Type A and Type B n-gram features with Spearman ρ and LOO retention fraction.

S3 Table. Top dating-driving features for the three Documentary Hypothesis source composites. For each source the three features most strongly indicating late composition and the three most strongly indicating early composition are listed with their normalized archaism scores and a note on potential genre confound. Scores follow Eq. 7: 0 = CBH-like, 1 = LBH-like, < 0 = more archaic than the CBH anchor, > 1 = more LBH-like than the LBH anchor. This table is provided to allow readers to evaluate the model's inferences directly, without relying on the fitted model weights.

S4 Table. Robustness check: MAP dates under progressive feature exclusion. The MLE-MVN model is refit after removing the two genre-flagged features identified in S3 Table (*asher* frequency for D; wayyiqtol fraction for JE) and after removing all features with normalized archaism scores exceeding $|s| > 2.5$ for any source. Feature vectors for D, P, and JE are reconstructed from normalized archaism scores (see Methods); absolute dates carry a minor approximation error from this reconstruction, so the shift columns (Δ) are the primary quantities of interest. All sources remain in the Persian-to-Hellenistic period under all exclusion schemes.

S5 Table. Corpus word counts for training texts and out-of-sample sources. All word counts are morphological words (BHSA) excluding coordinating conjunctions. The 19 training texts span 760–330 BCE and cover prophetic, poetic, narrative, and wisdom registers; three additional texts (Habakkuk, Haggai, Daniel) serve as cross-validation holdouts for the HB-VI model. Documentary Hypothesis source composites and sub-source divisions are excluded from both training and validation; their dates are derived solely from the model posterior.

Data availability

All feature extraction scripts, Bayesian dating models, prior sensitivity code, and corpus manifests are publicly available at `https://github.com/adairaar/diachronic-hebrew`. The underlying Hebrew text corpus is the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA), version 2021, available at `https://github.com/ETCBC/bhsa` under a Creative Commons Attribution 4.0 license [6]. The Ancient Greek corpus is drawn from the Perseus Digital Library (`http://www.perseus.tufts.edu`) and the Thesaurus Linguae Graecae (`http://stephanus.tlg.uci.edu`), both publicly accessible. No proprietary data were used.

Author contributions

A.A. conceived and designed the study, developed and implemented all computational methods, performed all analyses, interpreted the results, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Funding

The author received no specific funding for this work.

Ethics statement

This study does not involve human participants, vertebrate animals, or any data requiring ethical oversight. All textual data analyzed are publicly available historical corpora. No ethical approval was required. This study was not pre-registered; it is a methods paper reporting retrospective analysis of existing corpora.

Acknowledgments

The author thanks the ETCBC team for maintaining the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA) database and making it freely available.

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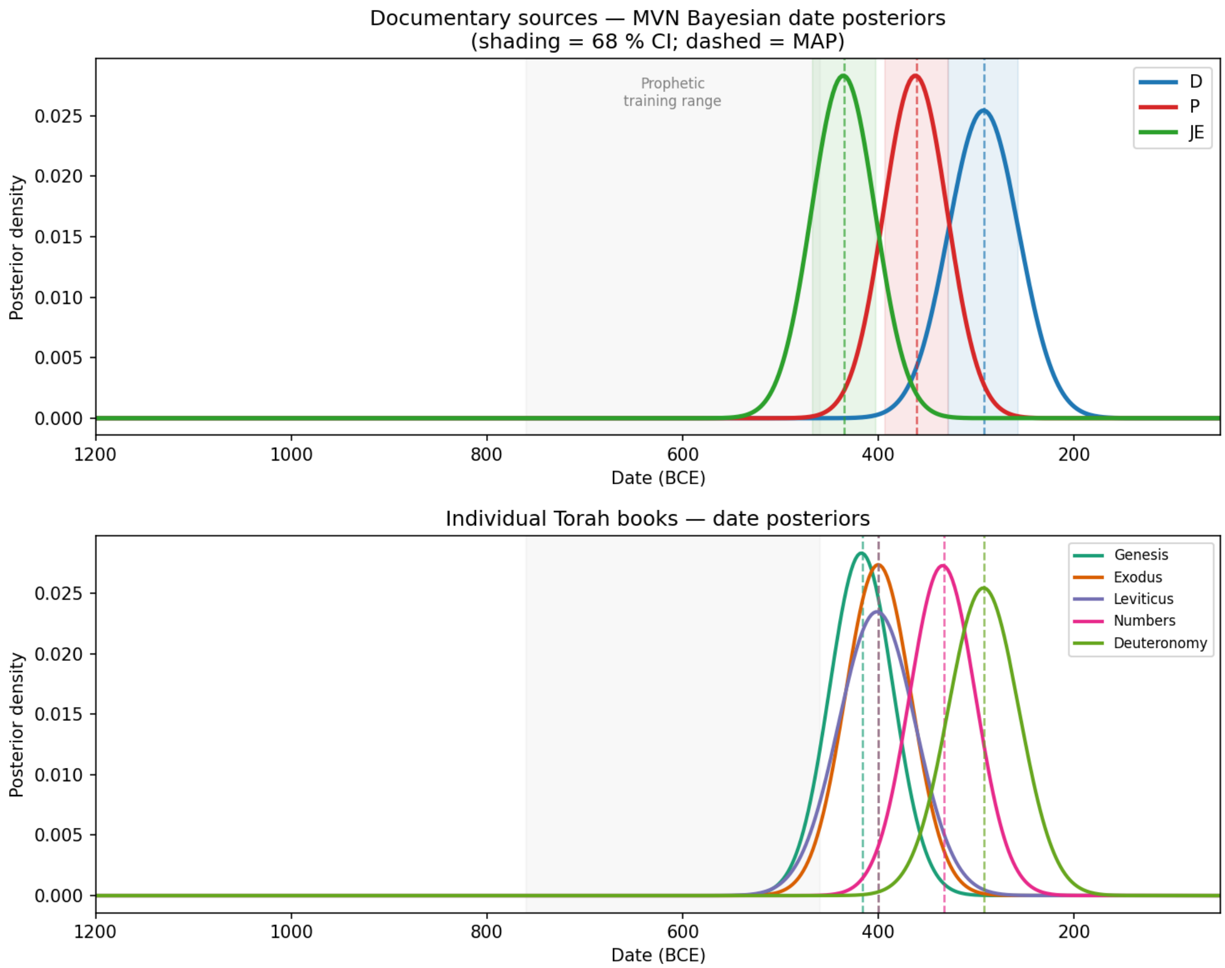

Fig 1. MLE-MVN date posteriors for Pentateuchal sources and books (broad agnostic prior). Top: full-model posteriors for the three Documentary Hypothesis source complexes treated as undated test texts (D = Deuteronomist, blue; P = Priestly, red; JE = Jahwist-Elohist combined, green). Bottom: posteriors for each of the five Pentateuchal books dated individually. Dashed vertical lines mark MAP dates; shaded bands indicate 68% HDCIs. The gray panel marks the prophetic training range (∼760–460 BCE). No posterior overlaps this range: all Torah material is placed in the Persian-to-Hellenistic period (∼480–250 BCE). The Priestly source is data-driven (prior sensitivity Class D, $|\Delta_{AB}| = 4$ yr); D and JE are prior-dominated (Class P, $|\Delta_{AB}| > 170$ yr; see Fig. 8 and Table 4). The training corpus and its register zones are shown in S12 Fig.

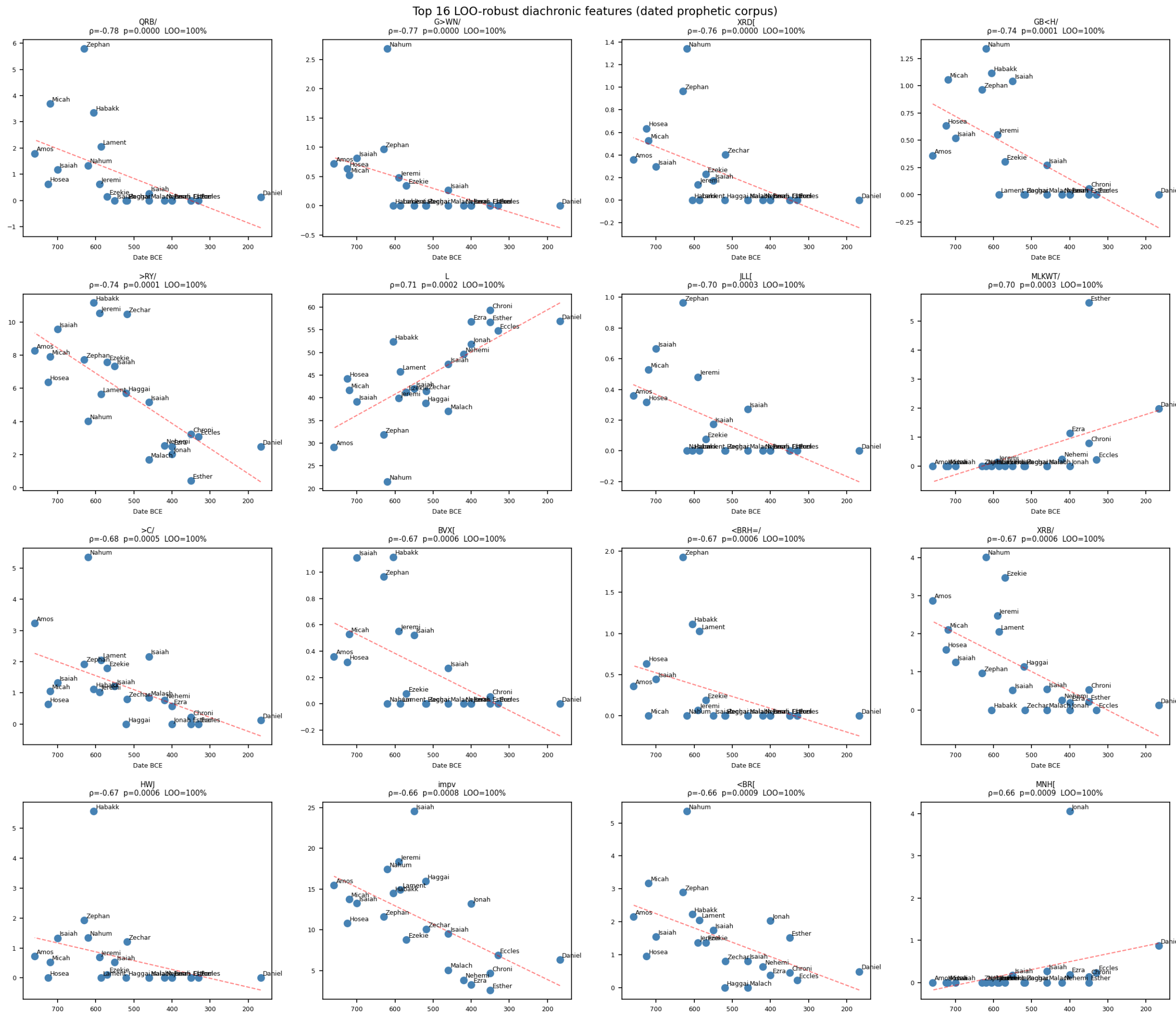

Fig 2. Top-10 morphosyntactic features by Spearman $|\rho|$ with date. Each panel shows the rate of one feature (y-axis) versus date BCE (x-axis) across the 22 training units, with OLS regression line and 95% confidence band. Features are color-coded by tier (Tier 1 = lexical, Tier 2 = morphological, Tier 3 = clause/phrase resistant). The *ankī/anī* ratio and wayyiqtol fraction show the steepest monotonic declines; infinitive-construct fraction (Tier 3) shows the most scatter but is retained as a resistant-model feature.

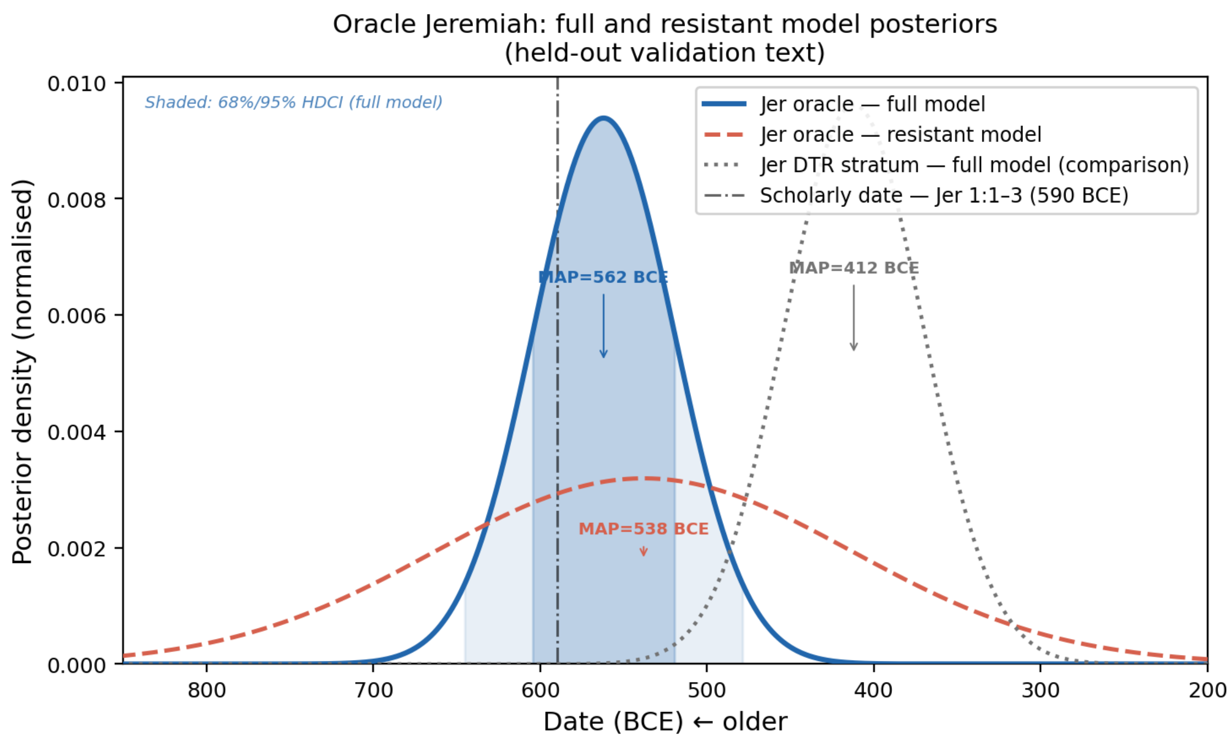

Fig 3. Oracle Jeremiah validation: full and resistant model posteriors. Both the full model (35 features) and the resistant model (4 Tier-3 features) are applied to the oracle Jeremiah chapters (Jer_oracle), which were excluded from training. Posteriors are shown as Gaussian approximations parameterised by the MVN-model MAP and 68% HDCI; shaded bands indicate the 68% and 95% regions for the full model. The full-model MAP (562 BCE) and the resistant-model MAP (538 BCE) agree within 24 yr ($\Delta_{\text{arch}} = +24$ yr), confirming the oracle chapters carry no archaizing signal and dating close to the scholarly consensus (≈ 590 BCE, dot-dash line). The Deuteronomistic prose stratum (Jer_DTR, dotted gray, MAP = 412 BCE) is shown for comparison; its distinctly later date is consistent with post-exilic Deuteronomistic expansion.

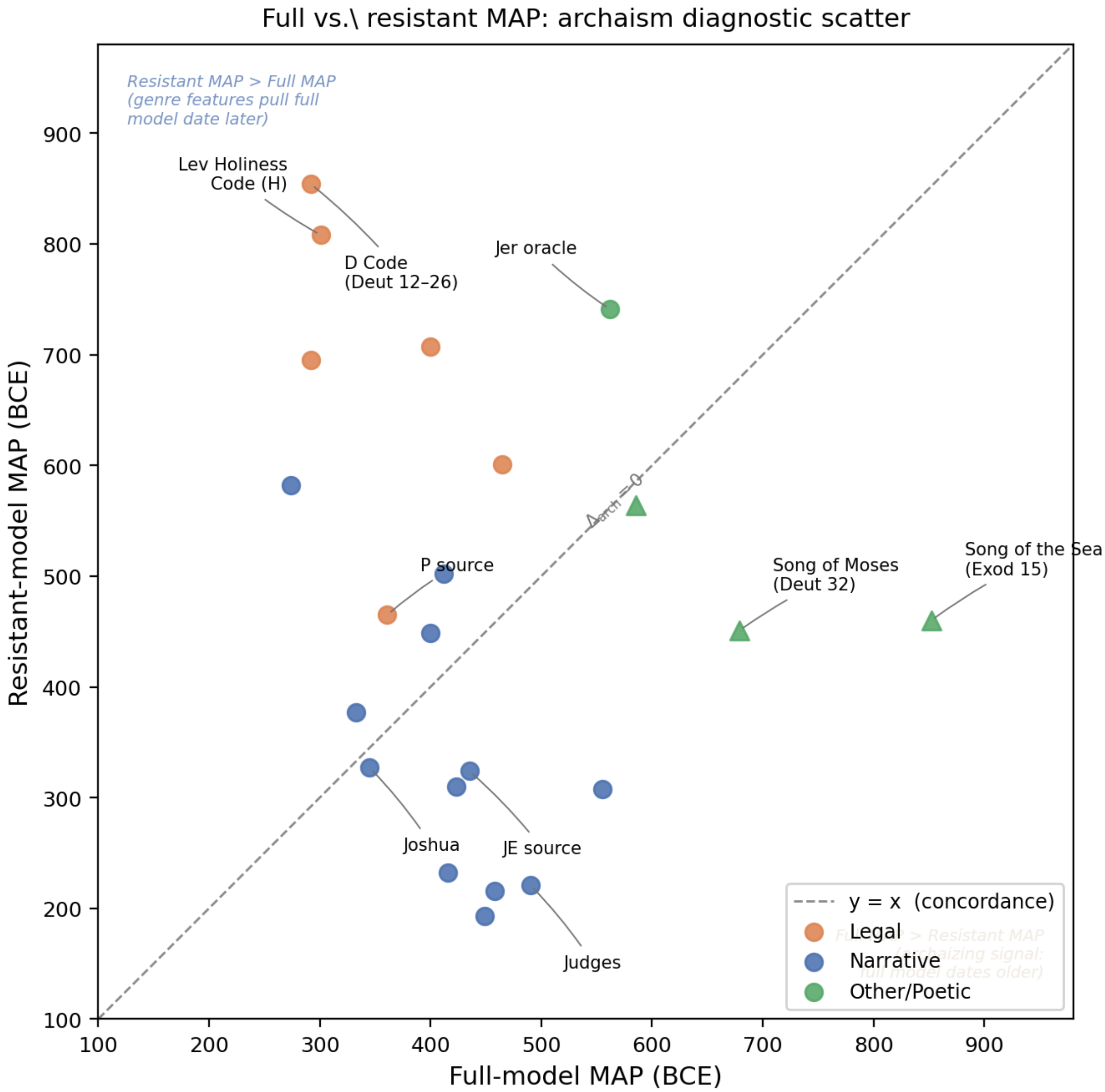

Fig 4. Full versus resistant MAP dates: archaism diagnostic scatter plot. Each of 22 analyzed textual units is plotted with full-model MAP (x-axis) versus resistant-model MAP (y-axis); filled circles = non-noisy texts, triangles = short or noisy texts. Colour indicates genre: blue = narrative, orange = legal, green = other/poetic. The diagonal dashed line indicates concordance ($\Delta_{\text{arch}} = 0$). Points *below* the diagonal (full MAP > resistant MAP, $\Delta_{\text{arch}} > 0$) signal archaizing composition—the full model dates the text more ancient than its clause-level syntax alone would suggest. Points *above* the diagonal ($\Delta_{\text{arch}} < 0$) indicate that genre features systematically pull the full-model date later than the underlying resistant syntax. Song of the Sea (Exod 15; $\Delta_{\text{arch}} = +392$ yr) and D_Code (Deut 12–26; $\Delta_{\text{arch}} = -562$ yr) are the most extreme positive and negative outliers, respectively.

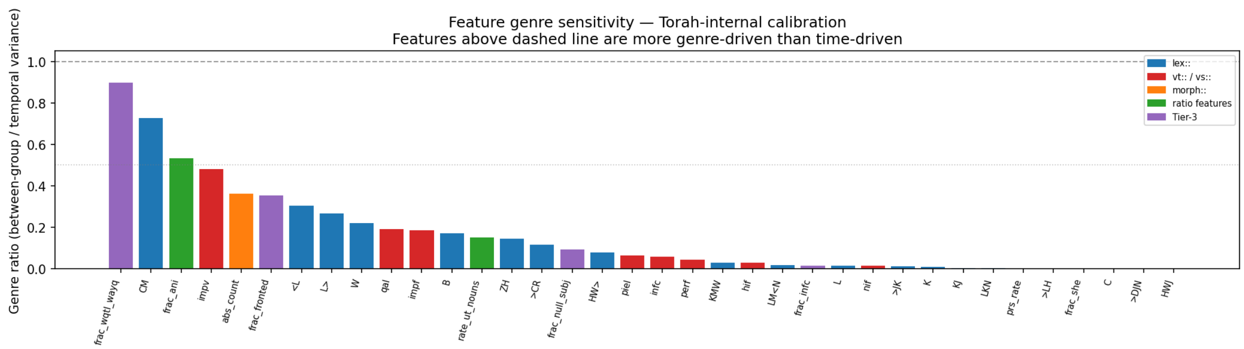

Fig 5. Genre sensitivity of the 35 morphosyntactic features. Features are ranked by genre ratio γ_j (between-genre variance as a fraction of temporal variance). Features to the right of the dashed threshold ($\gamma_j > 0.20$) are down-weighted in Strategy B genre correction. The 12 features selected for the genre-neutral sub-model are highlighted; these carry primarily temporal signal.

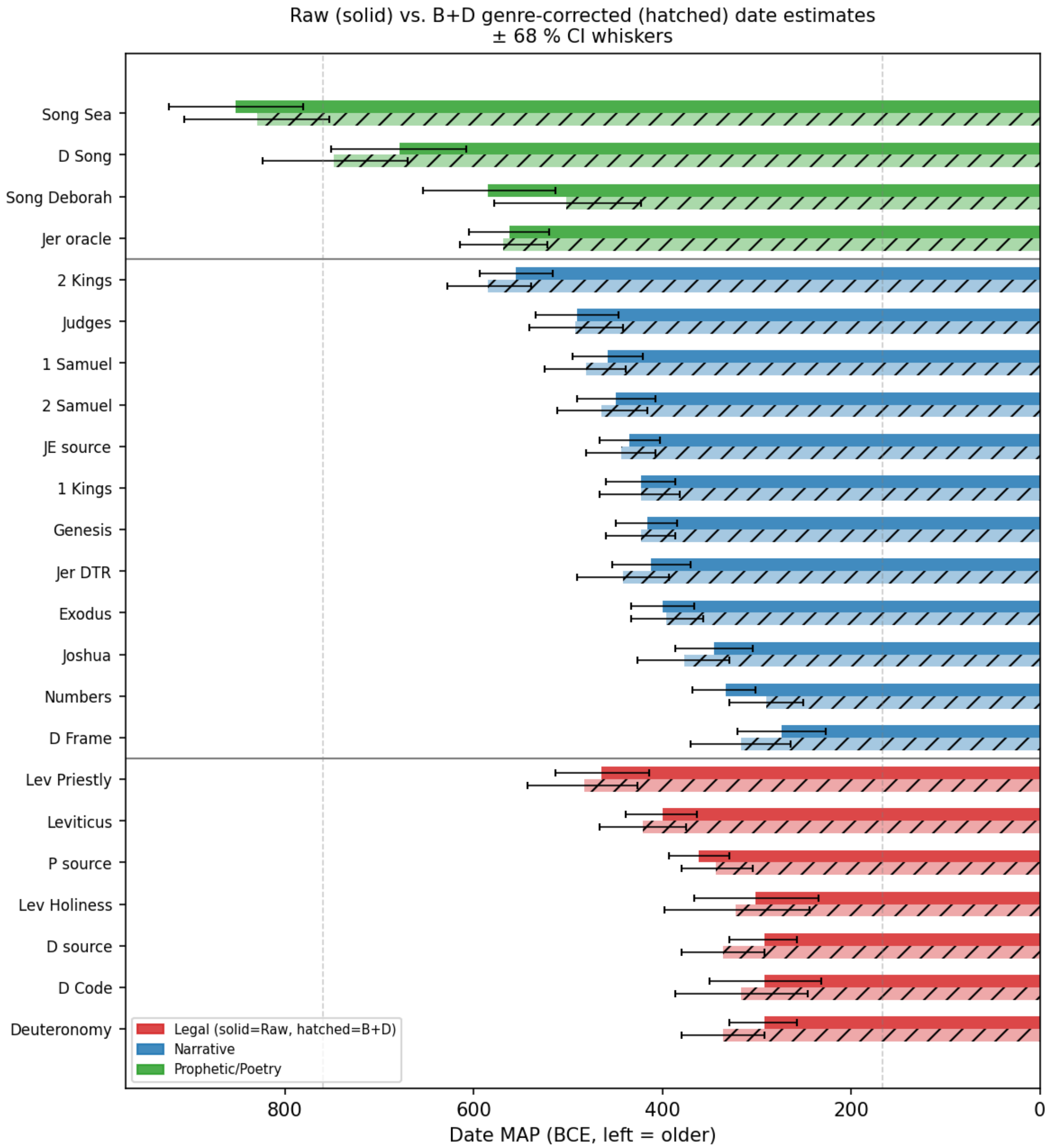

Fig 6. Genre correction (Strategy B+D): full model versus corrected MAP dates. Each bar shows the full-model MAP (gray) and B+D-corrected MAP (color) for Torah books and selected non-prophetic texts. Legal texts (Deuteronomy, Leviticus) shift older under correction; narrative texts (Genesis, Numbers) shift modestly; prophetic texts are near-zero. The rightmost panel compares genre-neutral MAP (open circles) to B+D MAP for legal texts, showing the large genre contamination for D and the convergence for P.

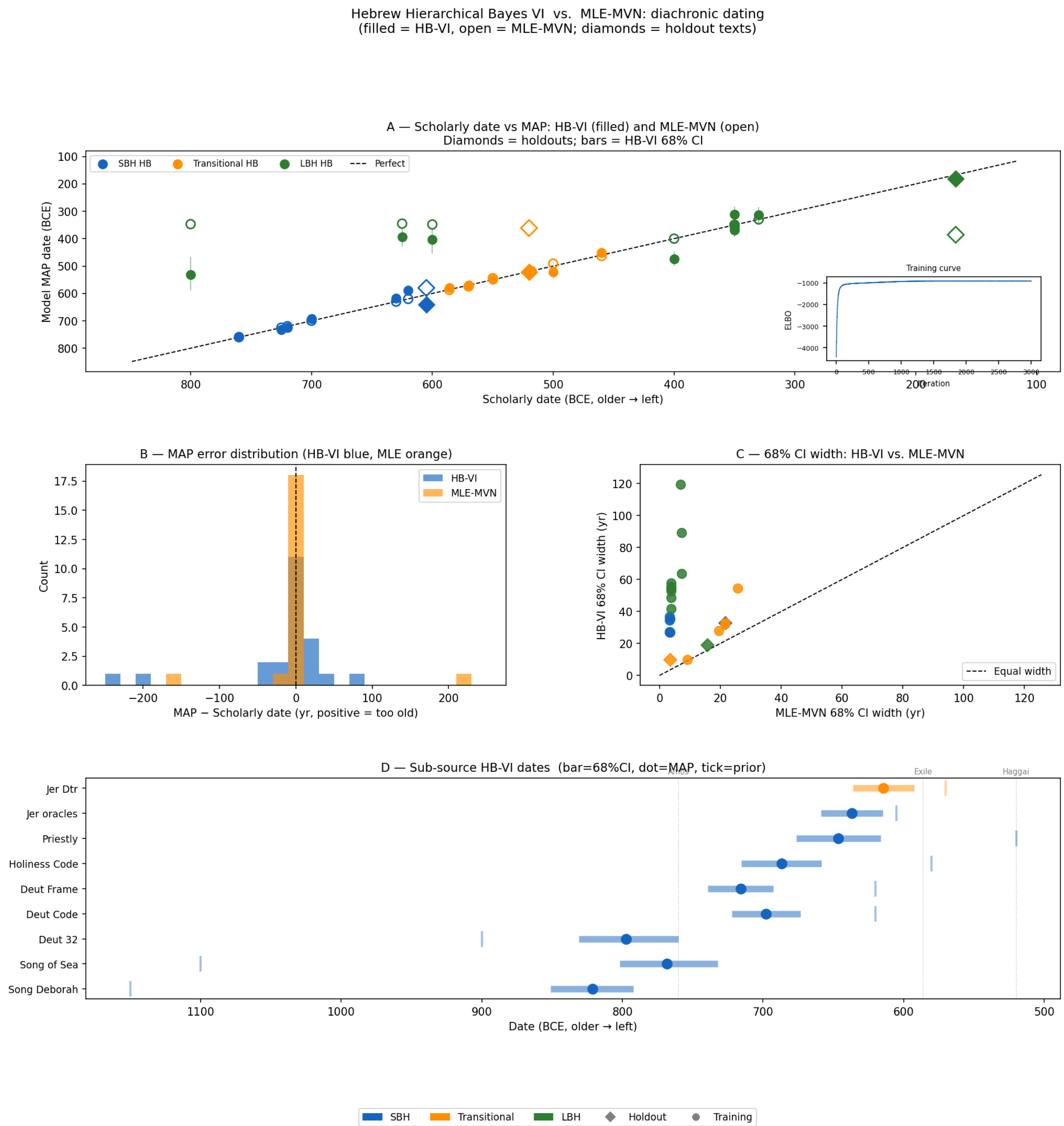

Fig 7. HB-VI versus MLE-MVN: MAP dates and 68% credible intervals for all training and comparison texts. Connected pairs of dots show MLE-MVN (triangles) and HB-VI (circles) MAP dates for each text; vertical bars indicate 68% HDCIs. The comparison texts (Habakkuk, Haggai, Daniel; shaded region) illustrate the key improvement: HB-VI correctly assigns Haggai to the Transitional group (MAP = 522 BCE) while MLE-MVN miscategorizes it as LBH (MAP = 361 BCE). HB-VI CIs are systematically wider and better-calibrated than MLE-MVN CIs.

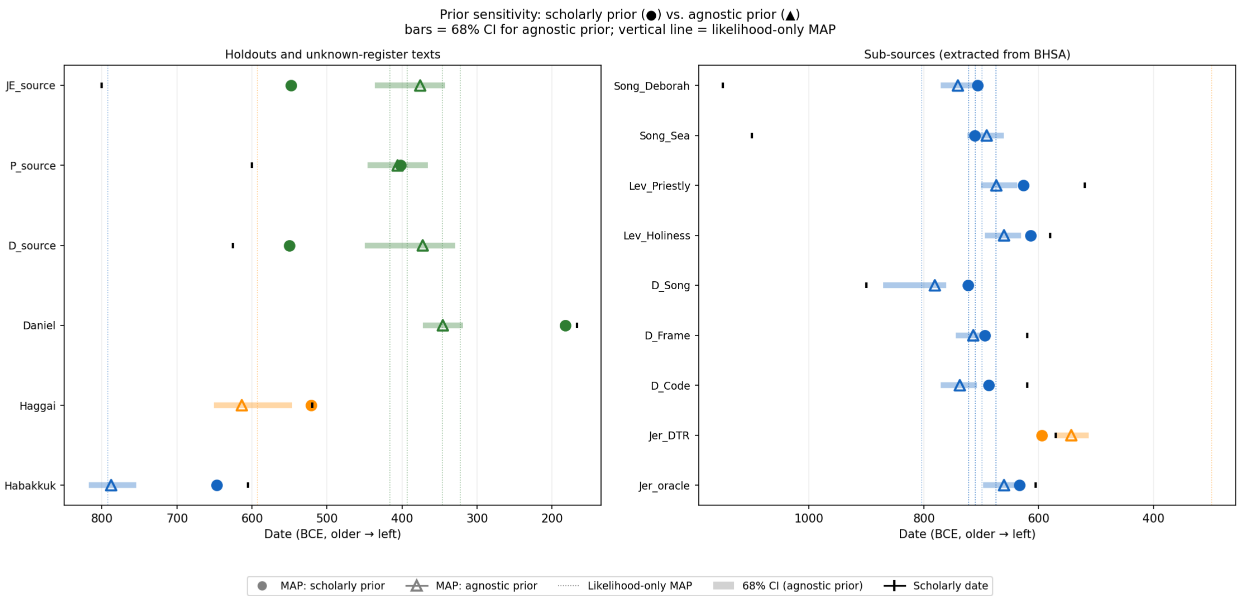

Fig 8. Prior sensitivity analysis: Mode A versus Mode B MAP dates for all sub-sources. Mode A (scholarly prior, circles) and Mode B (agnostic prior, triangles) MAP dates are connected for each text. Line length represents the Δ_{AB} shift; texts are colored by classification: green = data-driven ($|\Delta_{AB}|$<30 yr), orange = mildly prior-influenced (30–80 yr), red = prior-dominated ($>$ 80 yr). The four data-driven texts (Song_Sea, P_source, D_Frame, Jer_oracle) are annotated.

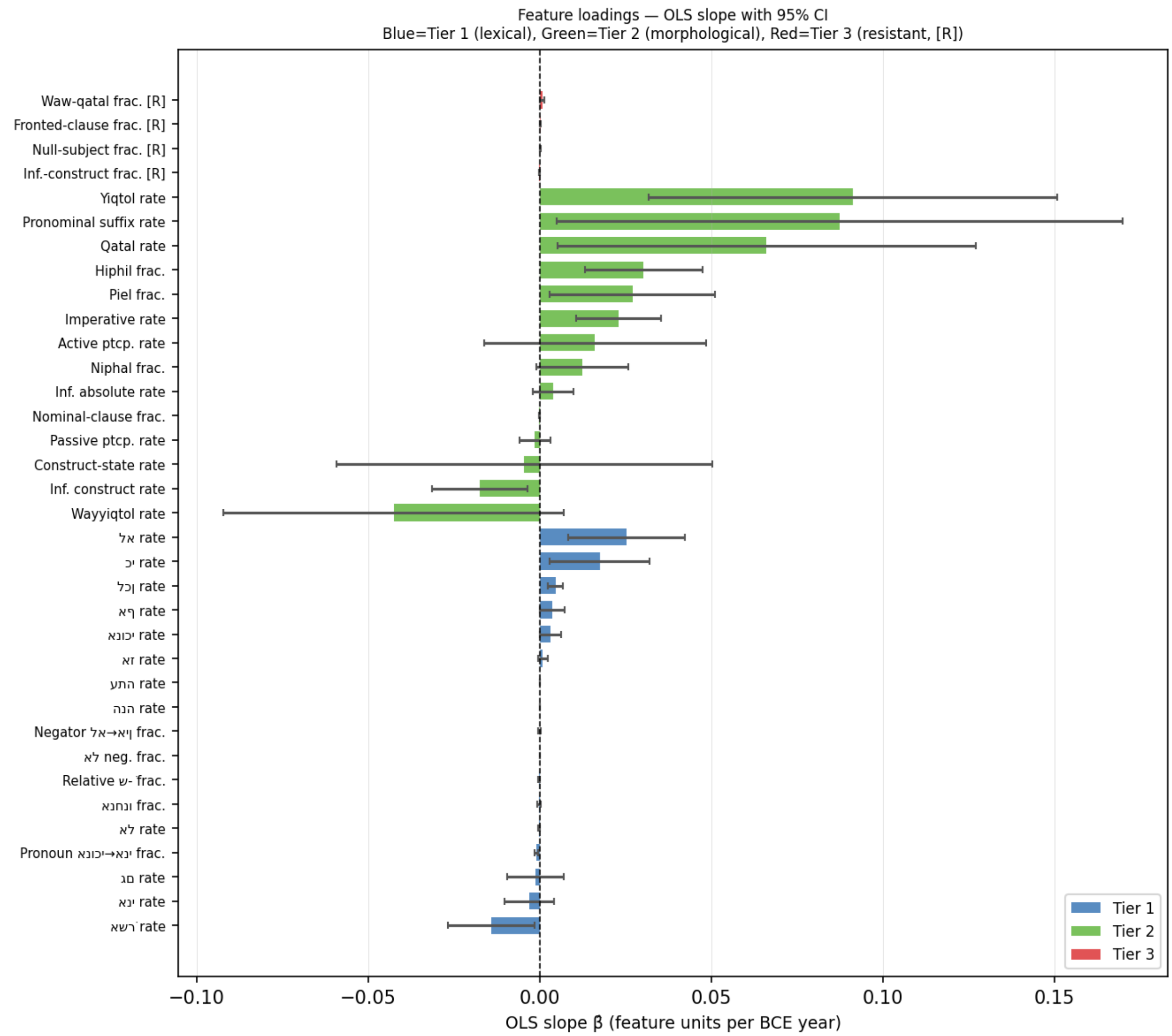

Fig 9. S6 Fig. Feature loadings (OLS slopes) for all 35 morphosyntactic features. $\hat{\beta}_j$ (OLS slope against date BCE) with 95% CI. Blue = Tier 1 (lexical), green = Tier 2 (morphological), red = Tier 3 (resistant, [R]). Negative slopes = feature declines with date; positive = rises with date.

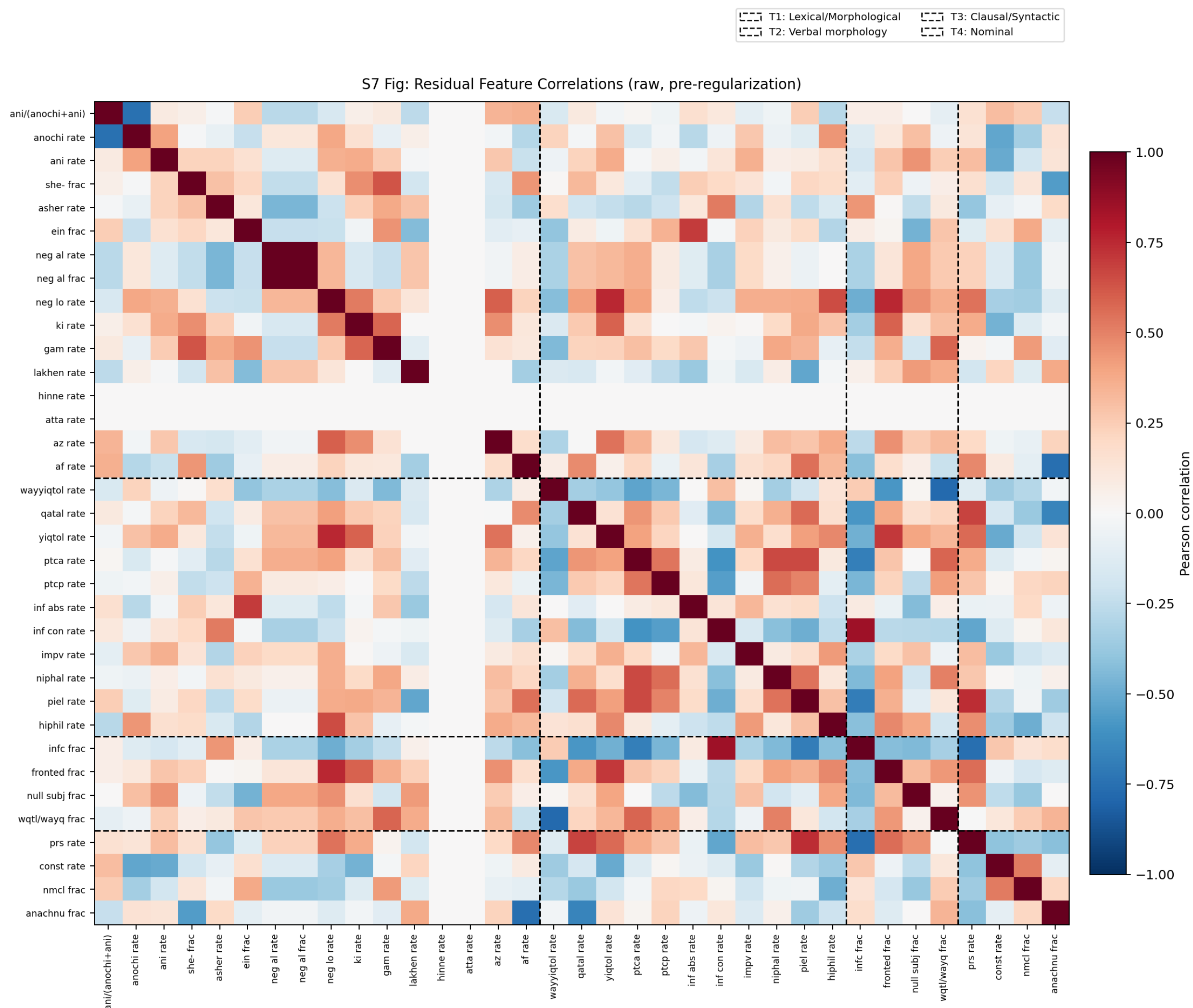

Fig 10. S7 Fig. Raw residual feature correlations (pre-regularisation). Pearson correlations among OLS residuals across the 22 training texts, computed *before* Tikhonov regularisation ($\lambda = 0.10$). Regularisation shrinks these off-diagonal entries toward zero to ensure the 35×35 covariance matrix $\mathbf{\Sigma}$ is invertible (the raw matrix is rank-deficient because $N = 22 < J = 35$). Tier boundaries are marked by dashed lines. Within-tier correlations—particularly the negative coupling between complementary forms (e.g., *wayyiqtol*/*qatal*, *asher*/*she-*)—confirm that features co-vary beyond the linear time trend, motivating the full MVN likelihood over an independent-feature approximation.

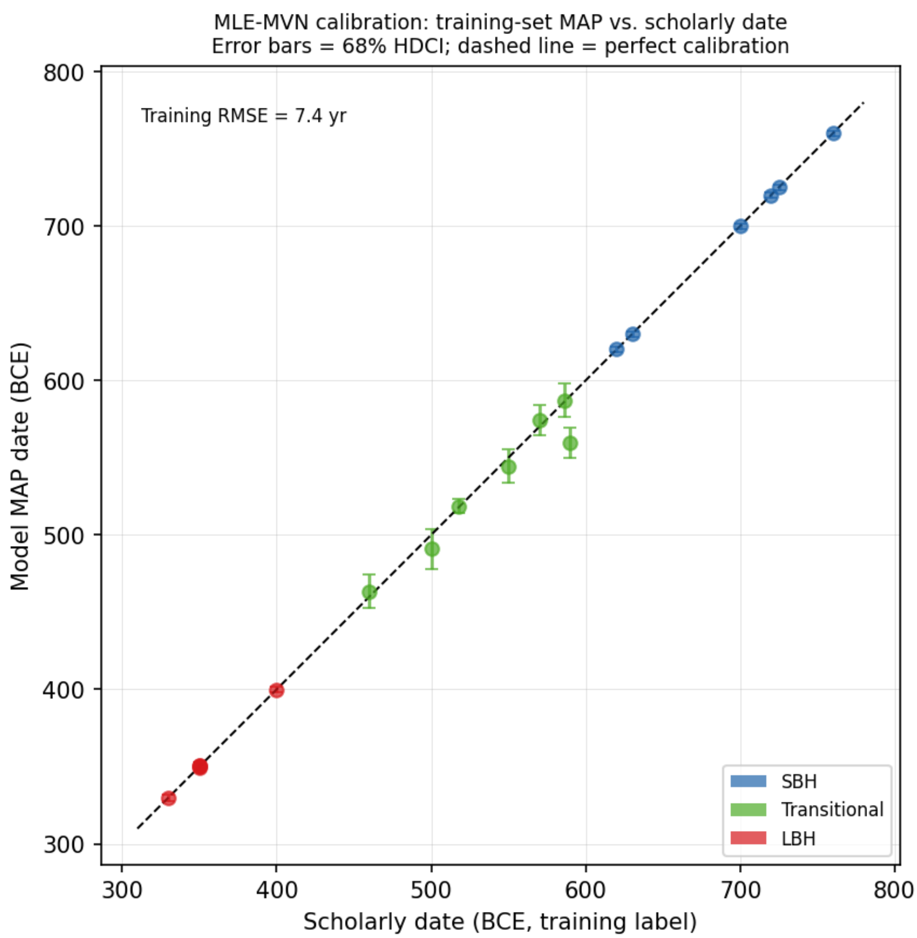

Fig 11. S8 Fig. MLE-MVN calibration: training-set MAP vs. scholarly date. Colors indicate register (blue = SBH, green = Transitional, red = LBH). Error bars = 68% HDCI. Dashed diagonal = perfect calibration.

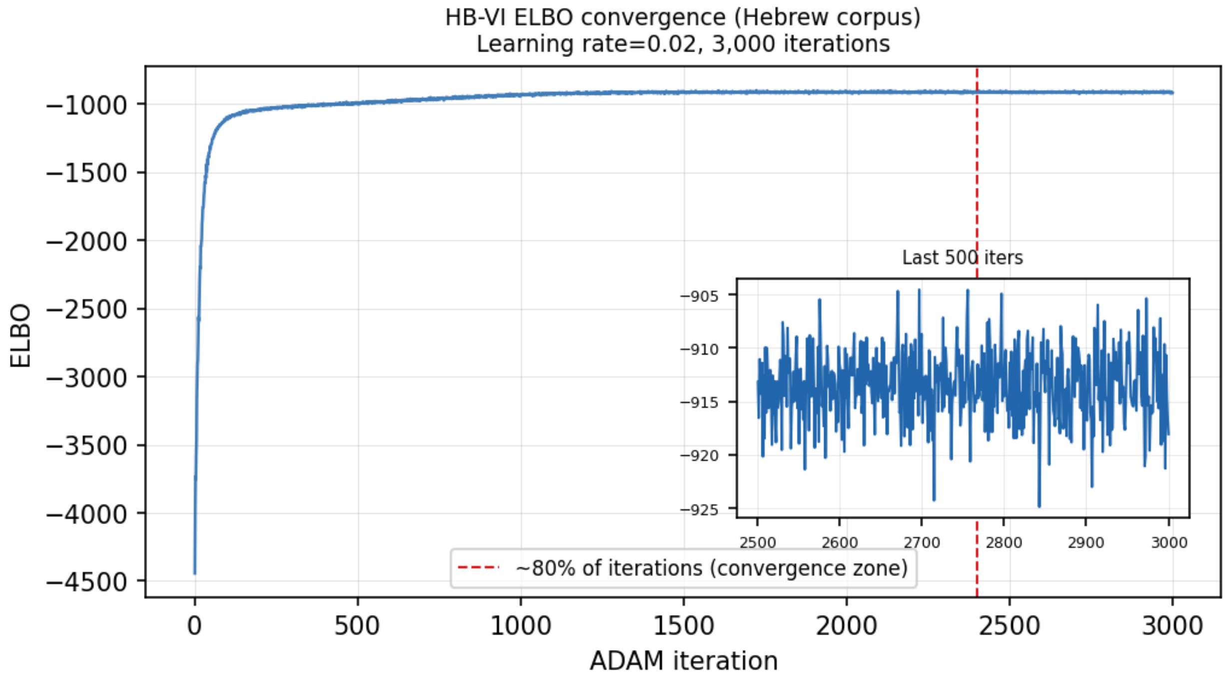

Fig 12. S9 Fig. HB-VI ELBO convergence. ELBO vs. ADAM iteration (learning rate = 0.02, 3,000 iterations). Dashed line marks the 80% convergence threshold; inset shows the final 500 iterations confirming near-flat behaviour.

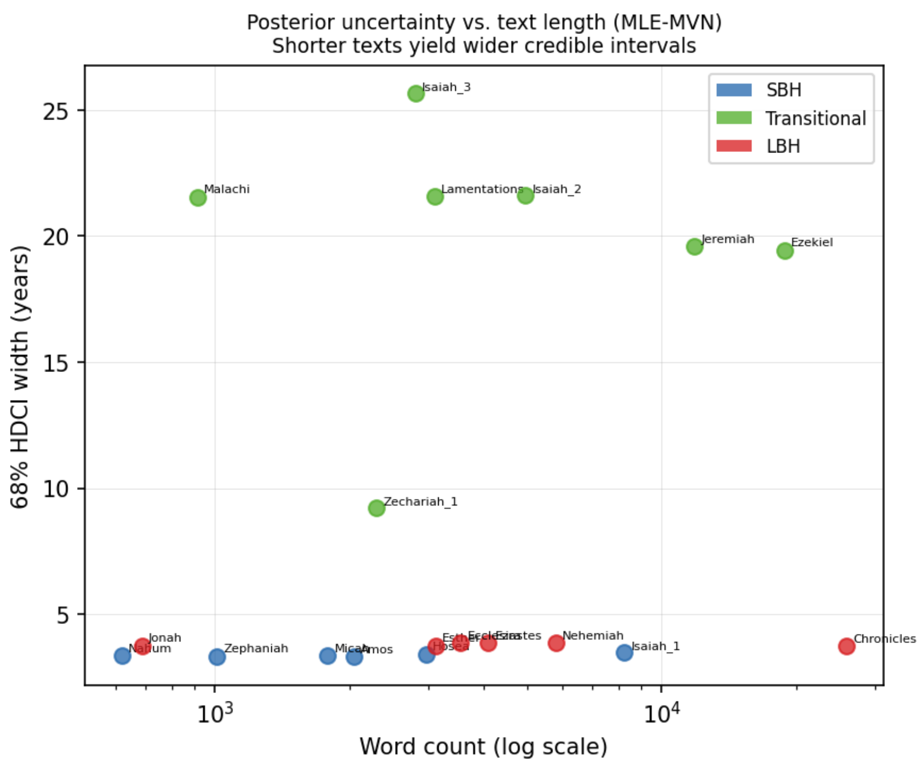

Fig 13. S10 Fig. 68% HDCI width vs. text length (MLE-MVN). Posterior uncertainty increases for shorter texts. Colors indicate register class.

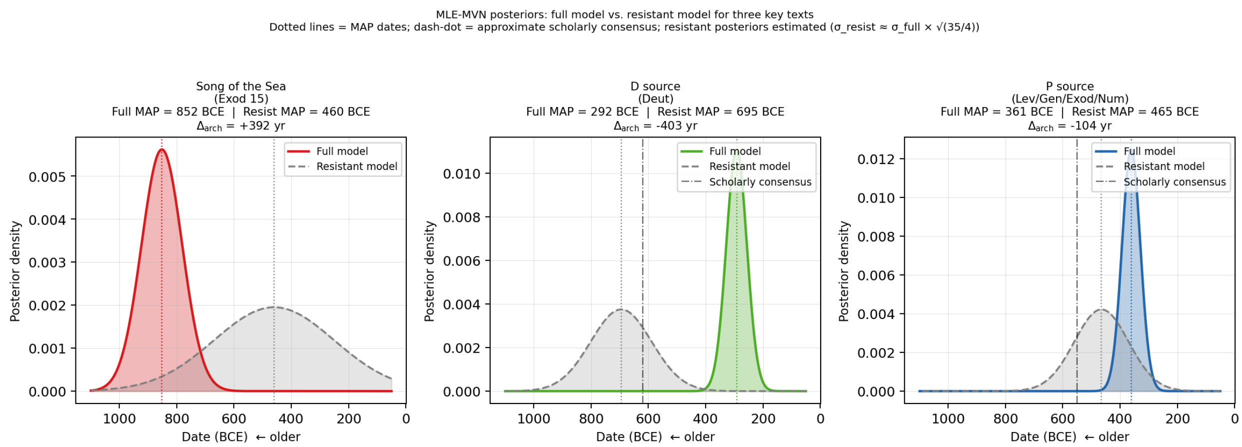

Fig 14. S11 Fig. Full MLE-MVN posteriors for three key texts. Full model (colored) vs. resistant model (gray dashed). Song of the Sea shows the ceiling effect and archaism diagnostic most clearly; the resistant-model MAP (≈460 BCE) diverges sharply from the full-model MAP (≈852 BCE), quantifying the archaism index $\Delta_{\mathrm{arch}} = +392$ yr.

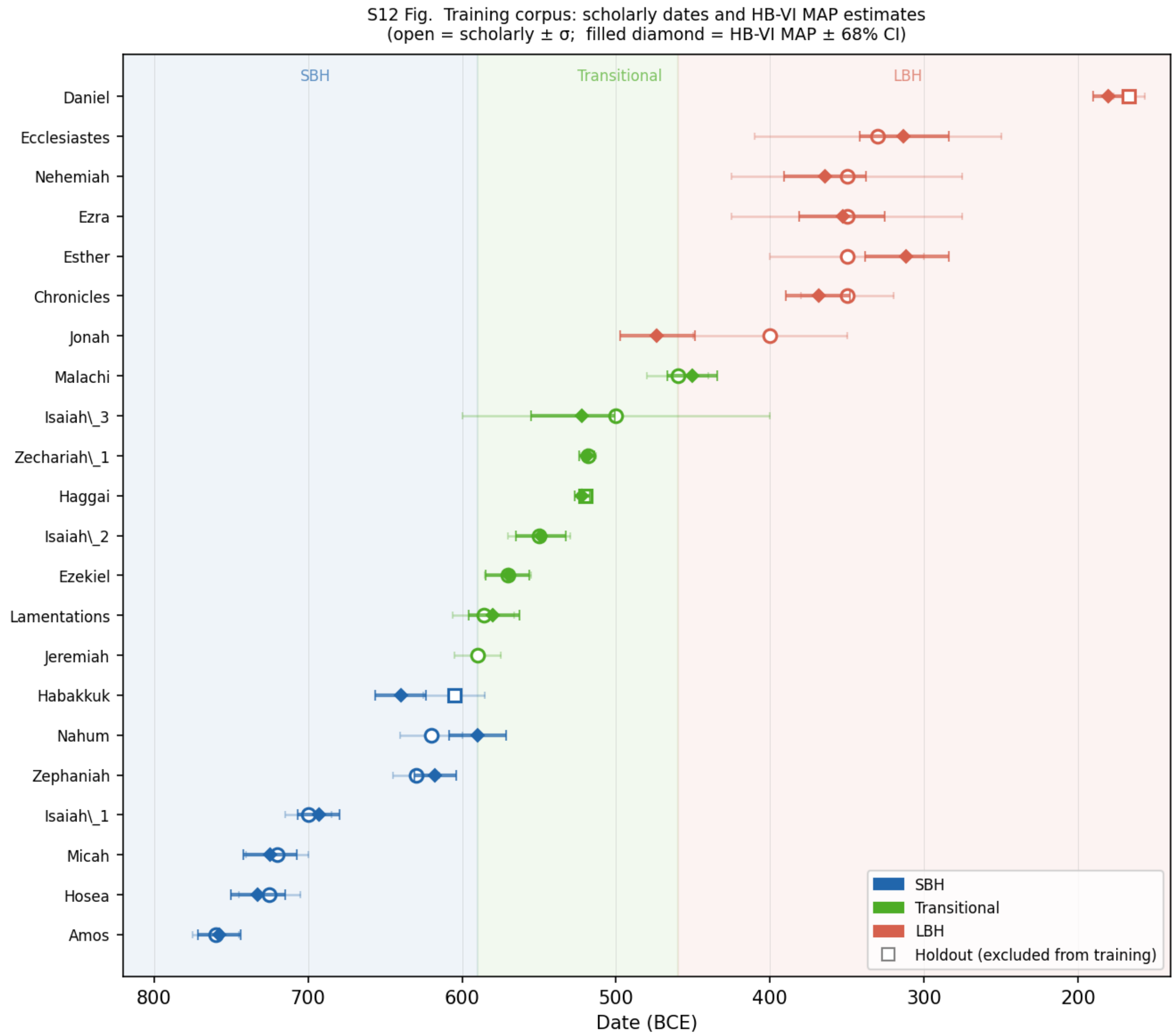

Fig 15. S12 Fig. Training corpus: texts, scholarly dates, and HB-VI MAP estimates. The 22 corpus texts (19 training, 3 holdouts: Habakkuk, Haggai, Daniel) are plotted on the BCE timeline. Open markers show scholarly consensus dates $\pm\sigma$; filled diamonds show HB-VI MAP estimates $\pm$ 68% HDCI. Color indicates register zone: blue = SBH (∼760–590 BCE), green = Transitional (∼590–460 BCE), red = LBH (∼460–167 BCE). The three holdouts were excluded from training and used solely for LOO cross-validation.

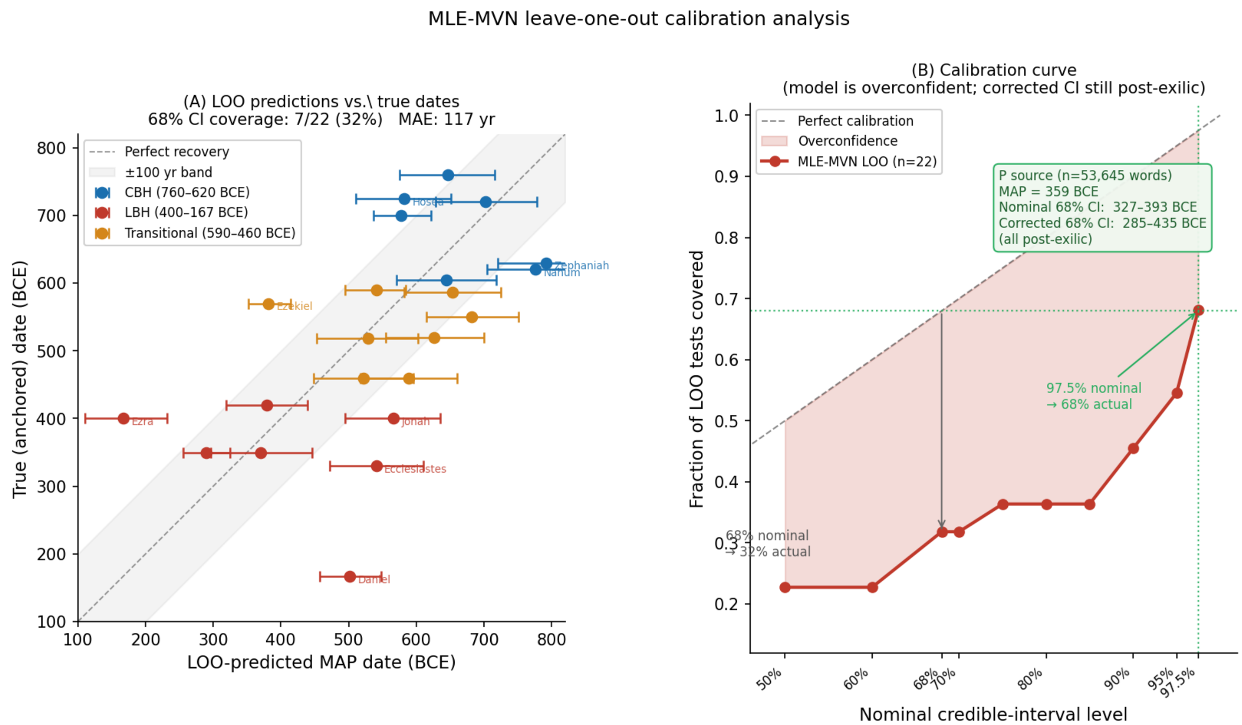

Fig 16. S13 Fig. Leave-one-out calibration analysis for the MLE-MVN model. (A) LOO-predicted MAP dates versus true (anchored) dates for all 22 training texts, with nominal 68% CI whiskers. Color indicates register: blue = CBH, orange = Transitional, red = LBH. Texts labeled in the plot are those with the largest LOO errors; all have contested or debated traditional dates in the scholarship. (B) Calibration curve showing actual LOO coverage (fraction of tests where the true date falls inside the CI) versus nominal credible-interval level. The shaded region marks the overconfidence gap. The nominal 68% CI achieves only 32% actual coverage; achieving 68% actual coverage requires expanding to the nominal 97.5% CI. The inset text shows that even under this conservative correction, the Priestly source (P; $n = 53{,}645$ words) has a calibration-corrected 68% interval of 285–435 BCE — entirely within the post-exilic period.

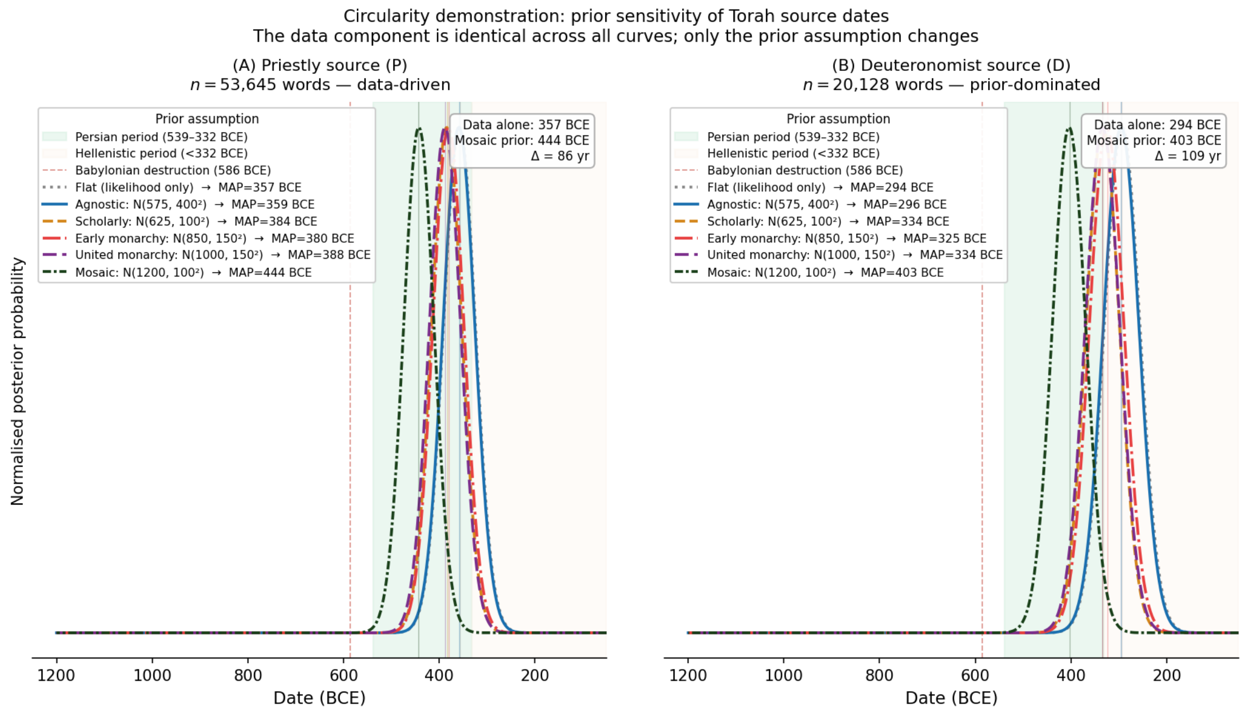

Fig 17. S14 Fig. Circularity demonstration: prior sensitivity of Torah source dates. Normalised MLE-MVN posterior curves for (A) the Priestly source ($n = 53{,}645$ words) and (B) the Deuteronomist source ($n = 20{,}128$ words), under six prior assumptions: a data-only flat prior, the agnostic prior $\mathcal{N}(575, 400^2)$, a scholarly prior $\mathcal{N}(625, 100^2)$, an early-monarchy prior $\mathcal{N}(850, 150^2)$, a united-monarchy prior $\mathcal{N}(1000, 150^2)$, and a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$. The green shading marks the Persian period (539–332 BCE); the vertical dashed red line marks the Babylonian destruction (586 BCE). Feature likelihoods are identical across all curves; only the prior assumption varies. Even the Mosaic prior yields MAP dates of 444 BCE (P) and 403 BCE (D), both firmly within the Persian period. The pre-exilic Torah date is not recoverable from the uncontaminated linguistic signal.

Table 6. S3 Table. Top dating-driving features for the three Documentary Hypothesis source composites. For each source the three features most strongly indicating late and early composition are shown with their normalized archaism scores (Eq. 7): $s = 0$ = CBH-like, $s = 1$ = LBH-like, $s < 0$ = more archaic than the SBH anchor, $s > 1$ = more LBH-like than the LBH anchor. "Genre flag" marks features whose value in this source is plausibly driven by prose genre rather than date (see text). All features are computed directly from BHSA morphological tags and can be verified without running the model.

Source	Direction	Feature	Score	Note
D source composite (Deuteronomist; $MAP_{HB\text{-}VI}$ = 698–737 BCE, MAP_{MVN} = 292 BCE)				
D	Late	*asher* relative particle rate	+3.31	Genre flag: high rate reflects legal-prose relative-clause density, not late date
D	Late	*eleh* "these" rate	+1.26	Consistent LBH lexical marker
D	Late	Infinitive-construct fraction	+1.25	LBH syntactic feature
D	Early	*anoki* vs. *ani* 1sg pronoun	−1.20	Classic SBH marker; archaic divine-speech convention in legal texts
D	Early	*sham* "there" rate	−1.27	Classic SBH marker
D	Early	*lema'an* "in order that" rate	−1.97	Classic SBH marker
P source composite (Priestly; $MAP_{HB\text{-}VI}$ = 402–406 BCE, data-driven Class D, $\lvert\Delta_{AB}\rvert = 4$ yr)				
P	Late	Niphal stem rate	+1.39	Consistent LBH morphological marker
P	Late	Abstract nominal rate	+1.32	Consistent LBH feature (high nominal density)
P	Late	*eleh* "these" rate	+1.28	Consistent LBH lexical marker
P	Early	Infinitive-construct rate	+0.15	Low inf. construct (legal/priestly prose pattern)
P	Early	*she-* relative particle rate	−0.06	Classic SBH marker; no late relative pronoun
P	Early	Construct-state noun count	−0.04	High construct ratio (SBH nominal style)
JE source composite (Jahwist-Elohist; $MAP_{HB\text{-}VI} \approx 435$ BCE, prior-dominated Class P, $\lvert\Delta_{AB}\rvert = 172$ yr)				
JE	Late	Wayyiqtol fraction of narrative verbs	+1.51	Genre flag: high rate reflects narrative genre, not exclusively date
JE	Late	*bet* preposition rate	+1.42	Consistent LBH prose marker
JE	Late	*waw* conjunction rate	+1.28	Consistent LBH narrative marker
JE	Early	*she-* relative particle rate	−0.05	Classic SBH marker; no late relative pronoun
JE	Early	*anoki* vs. *ani* 1sg pronoun	−0.46	SBH marker
JE	Early	*sham* "there" rate	−0.56	Classic SBH marker

Scores are computed as $s_j = (x_j - \hat{f}_j(720\,\text{BCE}))/(\hat{f}_j(250\,\text{BCE}) - \hat{f}_j(720\,\text{BCE}))$ where $\hat{f}_j$ is the OLS-fitted feature trajectory. Scores > 1 indicate the feature value exceeds the LBH anchor level. Features marked "Genre flag" should be interpreted with caution because the training corpus contains no legal-prose texts from the Persian or Hellenistic period; see text.

Table 7. S4 Table. Robustness of source MAP dates to exclusion of genre-confounded features. MLE-MVN MAP dates for D, P, and JE under progressive feature exclusion. R1 removes *asher* frequency (genre-flagged for D; see S3 Table). R2 additionally removes wayyiqtol fraction (genre-flagged for JE). R3 removes all features with normalized archaism score $|s| > 2.5$ for any source. Feature vectors are reconstructed from normalized archaism scores (see Methods); absolute dates carry a minor approximation error, so the shift column Δ is the primary quantity of interest. R2 and R3 produce results identical to R1, confirming that *asher* is the sole high-leverage outlier. All sources remain in the Persian-to-Hellenistic period under all exclusion schemes.

Source	**Baseline (BCE)**	**R1: drop *asher* (BCE)**	**Δ (yr)**	**R2 = R3 (BCE)**	**P**
D (Deuteronomist)	313	387	+74	387	P
P (Priestly)	347	368	+21	368	P
JE (Jahwist-Elohist)	442	456	+14	456	Persian–early Hell

Baseline dates use agnostic prior (Mode B, $\mathcal{N}(575, 400^2)$ BCE) to isolate the data signal from prior assumptions. Positive Δ indicates shift toward more recent date after feature exclusion. "Persian period" = 539–332 BCE; "Persian–early Hellenistic" = 539–300 BCE. The 74 yr shift for D after excluding *asher* confirms that the dominant late signal in D is genre-confounded; the remaining Persian-period date reflects the contribution of the remaining late lexical markers (*eleh*, infinitive construct; see S3 Table).

Table 8. S5 Table. Corpus word counts for training texts and out-of-sample sources. All word counts are morphological words (BHSA) excluding coordinating conjunctions. The 19 training texts span 760–330 BCE and cover prophetic, poetic, narrative, and wisdom registers; three additional texts (Habakkuk, Haggai, Daniel) serve as cross-validation holdouts for the HB-VI model. Documentary Hypothesis source composites and sub-source divisions are excluded from both training and validation; their dates are derived solely from the model posterior. Texts below ∼1,000 words produce noticeably wider posteriors due to sampling noise in feature estimates (see Fig. S10).

Text	**Date (BCE)**	**Genre**	**Words**	**Role**
(A) Training corpus and cross-validation holdouts (oldest to most recent)				
Amos	760 ± 15	Prophecy	2,780	Training
Hosea	725 ± 20	Prophecy	3,146	Training
Micah	720 ± 20	Prophecy	1,895	Training
Isaiah (chs. 1–39)	700 ± 15	Prophecy	13,504	Training
Zephaniah	630 ± 15	Prophecy	1,037	Training
Nahum	620 ± 20	Prophecy	746	Training
Habakkuk	605 ± 20	Prophecy	897	HB-VI holdout
Jeremiah	590 ± 15	Prophecy	14,539	Training
Lamentations	586 ± 20	Poetry	1,945	Training
Ezekiel	570 ± 15	Prophecy	26,182	Training
Deutero-Isaiah (chs. 40–55)	550 ± 20	Prophecy	5,738	Training
Haggai	520 ± 5	Prophecy	877	HB-VI holdout
Zechariah (chs. 1–8)	518 ± 5	Prophecy	2,484	Training
Trito-Isaiah (chs. 56–66)	500 ± 100	Prophecy	3,689	Training
Malachi	460 ± 20	Prophecy	1,187	Training
Jonah	400 ± 50	Narrative	985	Training
Chronicles	350 ± 30	Narrative	35,330	Training
Esther	350 ± 50	Narrative	4,621	Training
Ezra	350 ± 75	Narrative	5,268	Training
Nehemiah	350 ± 75	Narrative	7,842	Training
Ecclesiastes	330 ± 80	Wisdom	4,233	Training
Daniel	167 ± 10	Mixed	8,072	HB-VI holdout
(B) Out-of-sample Documentary Hypothesis sources (excluded from all model fitting)				
D source composite	—	Legal	20,128	Out-of-sample
P source composite	—	Legal	53,645	Out-of-sample
JE source composite	—	Narrative	40,867	Out-of-sample
D Code (Deut 12–26)	—	Legal	7,353	Out-of-sample
D Frame (Deut 1–11, 27–34)	—	Narrative	11,993	Out-of-sample
D Song (Deut 32–33)	—	Other	782	Out-of-sample
Lev Holiness Code (17–26)	—	Legal	5,958	Out-of-sample
Lev Priestly (1–16)	—	Legal	10,496	Out-of-sample

HB-VI holdouts (Habakkuk, Haggai, Daniel) are withheld from *all* model fitting; their MAP dates assess cross-validation accuracy only (Fig. 9). The training/holdout distinction applies to HB-VI; all 22 texts in panel (A) appear in MLE-MVN training. D Song (782 words) should be treated with caution given its small size and poetic genre. CBH = Classical Biblical Hebrew; LBH = Late Biblical Hebrew; Transitional = texts with mixed register features.

Table 9. S6 Table. Prior-sweep MAP dates for the three Torah source composites. MLE-MVN MAP dates (BCE) for the Priestly (P), Deuteronomist (D), and Yahwist-Elohist (JE) source composites under six prior scenarios ranging from an uninformative flat (data-only) prior to a Mosaic-authorship prior $\mathcal{N}(1200, 100^2)$. Feature vectors are held fixed across all rows (reconstructed from normalized archaism scores; see Methods and Fig. S14); only the prior assumption changes. The Babylonian destruction (586 BCE), the Cyrus decree (539 BCE), and the onset of the Hellenistic period (332 BCE) are shown for reference. None of the six scenarios recovers a pre-exilic MAP date for any source; the data signal is strong enough to anchor all three sources in the Persian period even under the most extreme traditional prior.

Prior scenario	$\boldsymbol{\mu_0}$ (BCE)	$\boldsymbol{\sigma_0}$ (yr)	**P MAP**	**D MAP**	**JE MAP**	**Period**
Flat (data only)	—	—	357	294	434	Persian
Agnostic $\mathcal{N}(575, 400^2)$	575	400	359	296	434	Persian
Scholarly $\mathcal{N}(625, 100^2)$	625	100	384	334	453	Persian
Early monarchy $\mathcal{N}(850, 150^2)$	850	150	380	325	453	Persian
United monarchy $\mathcal{N}(1000, 150^2)$	1000	150	388	334	461	Persian
Mosaic $\mathcal{N}(1200, 100^2)$	1200	100	444	403	511	Persian

Feature vectors reconstructed from normalized archaism scores (archaism_by_feature.csv) via OLS back-transform; see Methods. Word-count scaling clip$(n/5000, 1, 5)$ applied (P n = 53,645; D n = 20,128; JE n = 40,867 words). The P–Mosaic shift of 87 yr across the full prior range confirms the data-driven classification from the prior sensitivity analysis ($|\Delta_{AB}| = 4$ yr; Table 3).

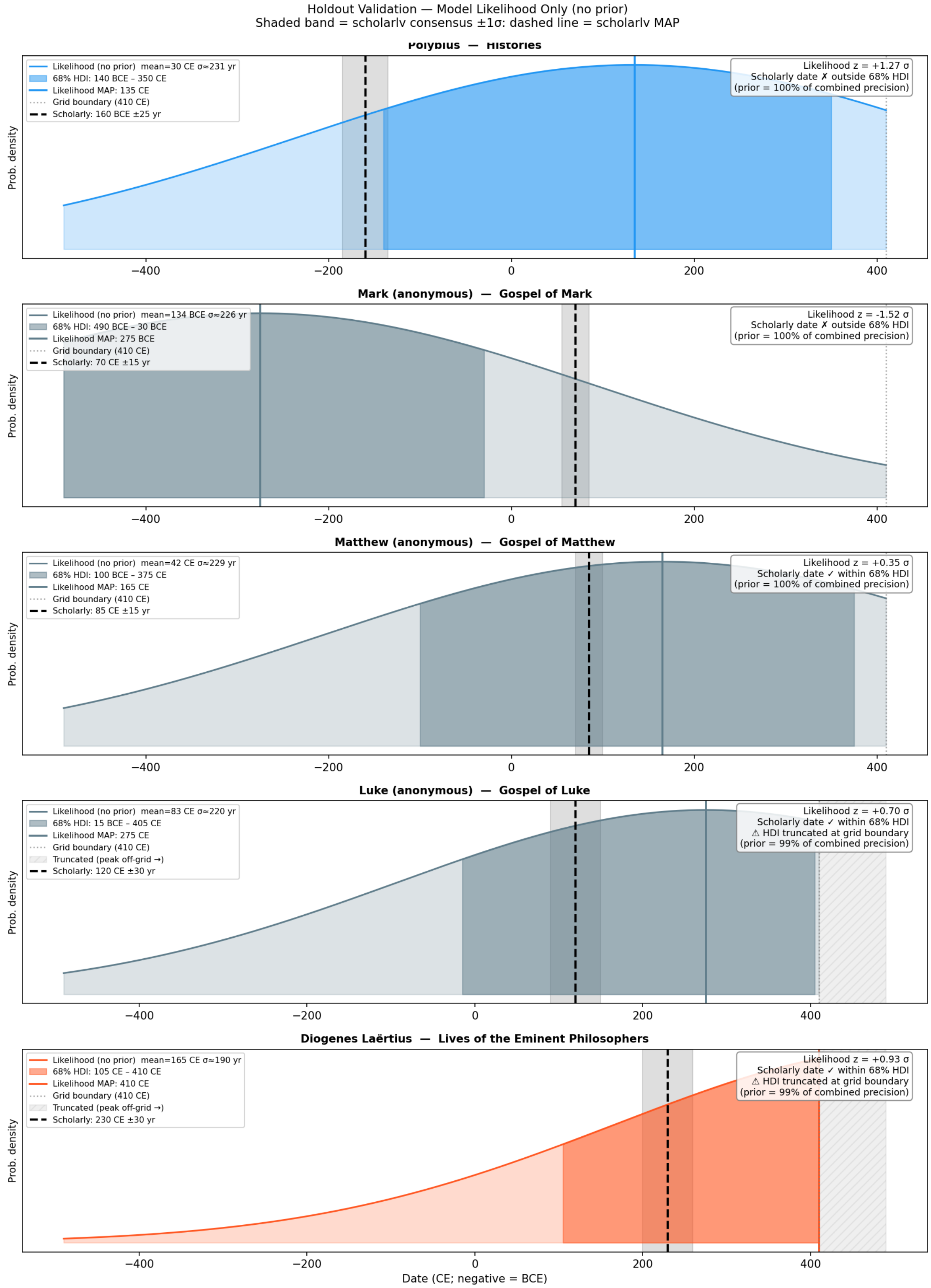

Fig 18. Cross-language validation on Ancient Greek (five holdouts). Each holdout is plotted at its independently established date (x-axis) versus the MLE-MVN MAP date (y-axis) with 68% CI whiskers. All five holdouts fall within ∼30 yr of the diagonal. Register class (ancient Attic, Koine, LXX, Atticizing) is indicated by symbol shape. The inset shows the LXX register diagnostic features for Mark and Luke, confirming the Semitic-substrate signal is correctly isolated from the temporal signal.